

An Efficient Minimax-Optimal Algorithm for Adversarial m -Set Bandits

Francesco Bacchiocchi¹, Tommaso Cesari², and Roberto Colomboni³

¹DEIB, Politecnico di Milano, Milano, Italy

²School of Electrical Engineering and Computer Science, University of Ottawa, Ottawa, Canada

³School of Mathematics, University of Bristol, Bristol, United Kingdom

francesco.bacchiocchi@polimi.it, tcesari@uottawa.ca, roberto.colomboni@bristol.ac.uk

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Abstract

We study adversarial combinatorial bandits with m -set actions, where at each round the learner selects m out of d items and observes only the aggregate loss of the selected items. The resulting action set contains $K = \binom{d}{m}$ elements and can therefore be exponentially large. Nevertheless, the loss of every action is determined by the same d -dimensional vector of item losses. We propose a computationally efficient algorithm that exploits this structure without explicitly enumerating the action set. Against adaptive non-anticipating adversaries, it guarantees, with probability at least $1 - \delta$, regret against the best fixed action of

$$R_T = O\left(\sqrt{dT \log(K/\delta)}\right).$$

This matches the high-probability regret bound of the finite-action EXP3-KW algorithm of Zimmert and Lattimore [31, Theorem 6 and Algorithm 3], whose direct implementation may require exponential space. Our algorithm instead represents each sampling distribution with d parameters and runs in polynomial time without enumerating the action set. Thus, it resolves the open problem posed by Maiti et al. [28]. We complement this upper bound with a matching high-probability lower bound. For all sufficiently small δ , every randomized policy admits a deterministic adaptive non-anticipating adversary for which, with probability at least δ ,

$$R_T = \Omega\left(\sqrt{dT \log(K/\delta)}\right).$$

Thus, the rate is minimax optimal up to universal constants in this regime. In particular, setting $m = 1$ proves that the $\log K$ for ordinary K -armed bandits against adaptive non-anticipating adversaries is unavoidable, closing the remaining $\sqrt{\log K}$ gap between confidence-tuned upper and lower bounds left by Gerchinovitz and Lattimore [16].

1 Introduction

In this paper, we study adversarial m -set bandits. At each round, the learner selects m out of d items and incurs the sum of their losses, but observes only the aggregate loss, not the loss of each selected item. Since any m -subset may be chosen, the learner has $K = \binom{d}{m}$ possible actions, and its goal is to minimize regret with respect to the best fixed m -set in hindsight.

For finite action sets, Zimmert and Lattimore [31, Theorem 6 and Algorithm 3] show that EXP3-KW achieves a high-probability regret bound that, for m -sets, becomes $\tilde{O}(\sqrt{dmT})$ when

$m \leq d/2$.¹ However, its direct implementation maintains one weight for every action, and since $K = \binom{d}{m}$, this can require exponential space. A possible workaround is to use the results in [28]. Specifically, Maiti et al. [28, Appendix E.6] give a polynomial-time high-probability algorithm based on a general directed acyclic graph representation that, for m -sets, turns into an algorithm achieving $\tilde{O}(d\sqrt{mT})$ regret bound. Despite this implementation being computationally efficient, it remains unclear whether one can efficiently obtain the sharper rate $\tilde{O}(\sqrt{dmT})$.

We provide a positive answer to this question with a polynomial-time algorithm. Against adaptive non-anticipating adversaries, our algorithm guarantees $\tilde{O}(\sqrt{dsT})$ regret with high probability, where $s := \min\{m, d - m\}$. In the regime $m \leq d/2$, we have $s = m$, and the bound becomes $\tilde{O}(\sqrt{dmT})$, and thus it matches the regret rate of finite-action EXP3-KW by Zimmert and Lattimore [31, Theorem 6 and Algorithm 3].

To obtain our efficient algorithm, we work with weighted m -set distributions, a family of distribution that can be represented by d item weights, from which one can sample and compute the required moments in polynomial time [12, 26]. The high-level idea is to implement the finite-action EXP3-KW algorithm while keeping the sampling distributions at all iterates within this family. The main obstacle is that the updates of EXP3-KW come with a quadratic correction, which does not preserve the weighted m -set structure. We overcome this difficulty by using the covariance bound of Cesari and Colomboni [11, Corollary 1.3] to construct a carefully chosen affine majorant of the quadratic correction on the m -set action space: exponential-weights updates using this majorant preserve the family of weighted m -set distributions, while the majorant also provides the estimation-error control needed to obtain the desired high-probability regret rate.

We also prove a matching high-probability lower bound for m -sets. For every d and m , for all sufficiently small δ , and for horizons above a polynomial threshold in d , m , and $\log(1/\delta)$, we construct a deterministic adaptive non-anticipating adversary for which, setting again $K = \binom{d}{m}$, it holds

$$R_T = \Omega\left(\sqrt{dT \log(K/\delta)}\right)$$

with probability at least δ . Thus, our upper bound is minimax optimal up to universal constants in this regime. This result also has an important consequence for standard adversarial multi-armed bandits. For K -armed bandits, the standard high-probability regret upper bound is $O(\sqrt{KT \log(K/\delta)})$ (see, e.g., [4]). In contrast, Gerchinovitz and Lattimore [16] prove a lower bound of order $\sqrt{KT \log(1/\delta)}$ even when the loss sequence is fixed in advance. Consequently, previously to this work, at constant confidence, that is, when δ is fixed, the known upper and lower bounds differed by a factor of $\sqrt{\log K}$. Setting $m = 1$ and $d = K$ in our lower bound shows that, against adaptive non-anticipating adversaries, every randomized K -armed bandit policy incurs regret at least

$$\Omega\left(\sqrt{KT \log(K/\delta)}\right)$$

with probability at least δ , for all sufficiently large horizons T . This matches the standard high-probability upper bound up to universal constants and therefore closes the remaining $\sqrt{\log K}$ gap in this adversary model.

¹The notation $\tilde{O}$ suppresses logarithmic factors.

2 Setting and problem formulation

Preliminary definitions. For a positive integer n , we write $[n] := \{1, \dots, n\}$. For a vector $z = (z_1, \dots, z_n)$ and $k \in [n]$, we define its k -th elementary symmetric polynomial as

$$e_k(z) := \sum_{S \subseteq [n]: |S|=k} \prod_{i \in S} z_i, \quad e_0(z) := 1, \quad e_k(z) := 0 \quad \text{for } k < 0 \text{ or } k > n.$$

Fix two integers $d \geq 2$ and $1 \leq m \leq d-1$. We define the action set and its size as

$$\mathcal{X} := \mathcal{X}_{d,m} := \left\{ x \in \{0, 1\}^d : \|x\|_1 = m \right\}, \quad K := |\mathcal{X}| = \binom{d}{m}. \quad (1)$$

Each action $x \in \mathcal{X}$ selects exactly m of the d items, $x_i = 1$ if item i is selected and $x_i = 0$ otherwise. Its support is $\text{supp}(x) := \{i \in [d] : x_i = 1\}$. Thus, the support of each action is an m -element subset of $[d]$, which we call an m -set. Conversely, each m -set $S \subseteq [d]$ corresponds to the indicator vector $\mathbf{1}_S \in \mathcal{X}$. We therefore use actions in $\mathcal{X}$ and m -sets interchangeably.

Let $s := \min\{m, d-m\}$. By the symmetry of the binomial coefficient, $\log K \leq s \log(ed/s)$.

Linear-algebra notation. For a symmetric matrix $A \in \mathbb{R}^{d \times d}$, we denote its kernel by $\ker(A) := \{z \in \mathbb{R}^d : Az = 0\}$ and its Moore–Penrose pseudoinverse by A^+ . For two symmetric matrices $A, B \in \mathbb{R}^{d \times d}$, we write $A \succeq B$ if $A - B$ is positive semidefinite or, equivalently

$$z^\top Az \geq z^\top Bz \quad \text{for every } z \in \mathbb{R}^d.$$

We write $A \succ 0$ if $z^\top Az > 0$ for every nonzero $z \in \mathbb{R}^d$. In particular, A is positive semidefinite if $A \succeq 0$, and positive definite if $A \succ 0$. Let $\mathbf{1} := (1, \dots, 1)^\top$, and define

$$\text{span}\{\mathbf{1}\} := \{a\mathbf{1} : a \in \mathbb{R}\}, \quad H_0 := \{y \in \mathbb{R}^d : \langle y, \mathbf{1} \rangle = 0\}.$$

Thus, H_0 is the subspace orthogonal to $\text{span}\{\mathbf{1}\}$. If A is positive semidefinite and $\ker(A) = \text{span}\{\mathbf{1}\}$, then A is invertible on H_0 . For every $y \in H_0$, the vector A^+y is the inverse of A applied to y within this subspace.

We will use the order-reversing property of matrix inversion [20, Corollary 7.7.4(a)]. Specifically, suppose that A and B are positive semidefinite linear operator with

$$\ker(A) = \ker(B) = \text{span}\{\mathbf{1}\},$$

then their restrictions to $H_0 = \mathbf{1}^\perp$ are positive definite. Consequently, if $A \succeq cB$, for $c > 0$, then

$$(A|_{H_0})^{-1} \preceq \frac{1}{c} (B|_{H_0})^{-1}.$$

Equivalently,

$$A^+ \preceq \frac{1}{c} B^+.$$

In particular,

$$y^\top A^+ y \leq \frac{1}{c} y^\top B^+ y \quad \text{for every } y \in H_0.$$

Probability notation. For a finite set $\mathcal{Y} \subseteq \mathbb{R}^d$, we denote its probability simplex by

$$\Delta(\mathcal{Y}) := \left\{ p : \mathcal{Y} \rightarrow [0, 1] : \sum_{y \in \mathcal{Y}} p(y) = 1 \right\}.$$

For $p, q \in \Delta(\mathcal{Y})$, we define the entropy of p and the Kullback–Leibler (KL) divergence from p to q as

$$H(p) := - \sum_{y \in \mathcal{Y}} p(y) \log p(y), \quad \text{KL}(p \parallel q) := \sum_{y \in \mathcal{Y}} p(y) \log \frac{p(y)}{q(y)}.$$

We use the convention $0 \log(0/q) = 0$, and set $\text{KL}(p \parallel q) := +\infty$ if $p(y) > 0 = q(y)$ for some $y \in \mathcal{Y}$.

Let X be a random vector drawn from $p \in \Delta(\mathcal{Y})$. We define the marginal vector, second-moment matrix, and covariance matrix of p as

$$\mu(p) := \mathbb{E}_{X \sim p}[X], \quad M(p) := \mathbb{E}_{X \sim p}[XX^\top], \quad \Sigma(p) := \text{Cov}_{X \sim p}(X) = M(p) - \mu(p)\mu(p)^\top.$$

When $\mathcal{Y} = \mathcal{X}$, the i -th marginal probability is

$$\mu_i(p) = \mathbb{E}_{X \sim p}[X_i] = \mathbb{P}_{X \sim p}[X_i = 1].$$

Thus, $\mu_i(p)$ is the probability that action X selects item i . If $M(p)$ is invertible, we refer to the inverse-design quadratic form

$$x^\top M(p)^{-1} x$$

as the *leverage score* of x under p .

Weighted m -set distributions. Given $\theta \in \mathbb{R}^d$, define $w_i := e^{\theta_i} > 0$ for every $i \in [d]$. For every m -set $S \subseteq [d]$, the weighted m -set distribution p_θ is

$$p_\theta(S) := \frac{\exp\left(\sum_{i \in S} \theta_i\right)}{\sum_{A \subseteq [d]: |A|=m} \exp\left(\sum_{i \in A} \theta_i\right)} = \frac{\prod_{i \in S} w_i}{e_m(w)}.$$

Its log-partition function is

$$F(\theta) := \log \sum_{A \subseteq [d]: |A|=m} \exp\left(\sum_{i \in A} \theta_i\right) = \log e_m(e^{\theta_1}, \dots, e^{\theta_d}).$$

If $S \sim p_\theta$ and $X := \mathbf{1}_S \in \mathcal{X}$ is its indicator vector, then

$$\nabla F(\theta) = \mathbb{E}_{X \sim p_\theta}[X] = \mu(p_\theta), \quad \nabla^2 F(\theta) = \text{Cov}_{X \sim p_\theta}(X) = \Sigma(p_\theta).$$

Although p_θ may assign positive probability to exponentially many actions, the d parameters $\theta_1, \dots, \theta_d$ represent it. Finally, we define a set of distributions whose marginal probabilities stay away from 0 and 1. Let $r := m/d$ and fix $\lambda \in (0, 1)$. Define

$$\mathcal{P}_\lambda := \{p \in \Delta(\mathcal{X}) : \lambda r \leq \mu_i(p) \leq 1 - \lambda(1 - r) \text{ for every } i \in [d]\}. \quad (2)$$

The uniform distribution U over $\mathcal{X}$ satisfies $\mu_i(U) = r$ for every $i \in [d]$. Therefore, $U \in \mathcal{P}_\lambda$.

Adversarial m -set bandit protocol. Fix a time horizon $T \geq 1$. Let $(\mathcal{F}_t)_{t=0}^T$ denote the filtration generated by the interaction history, where $\mathcal{F}_{t-1}$ contains the information available before round t . At each round $t \in [T]$, the learner selects an $\mathcal{F}_{t-1}$ -measurable distribution $p_t \in \Delta(\mathcal{X})$, where $\mathcal{X}$, defined in Equation (1), represents the set of actions available to the learner. The adversary selects a loss vector $\ell_t \in \mathbb{R}^d$ that is also $\mathcal{F}_{t-1}$ -measurable. Thus, the adversary is *adaptive non-anticipating*. The loss vector ℓ_t may depend on the entire interaction history up to round $t-1$, but not on the learner's random draw at round t . Therefore, conditionally on $\mathcal{F}_{t-1}$, both p_t and ℓ_t are fixed. The learner then draws $X_t \sim p_t$ and observes only the loss $Y_t := \langle X_t, \ell_t \rangle$. We use the standard action-level normalization

$$|\langle x, \ell_t \rangle| \leq 1 \quad \forall x \in \mathcal{X}, t \in [T],$$

also adopted by Maiti et al. [28, Assumption 1] and Zimmert and Lattimore [31, Section 1].² The learner's realized regret after T rounds, measured against the best fixed action in hindsight, is

$$R_T := \sum_{t=1}^T \langle X_t, \ell_t \rangle - \min_{x \in \mathcal{X}} \sum_{t=1}^T \langle x, \ell_t \rangle.$$

Throughout, we use the shorthand $\mathbb{E}_t[\cdot] := \mathbb{E}[\cdot | \mathcal{F}_{t-1}]$ for conditional expectation given the information available before round $t \in [T]$.

3 Original contributions

Our first contribution is an algorithm (Algorithm 1) for adversarial m -set bandits that achieves the regret guarantee summarized in the following.

Theorem (Informal version of Theorem 1). *There exists an algorithm that can be implemented in polynomial time such that, against any adaptive non-anticipating adversary, with probability at least $1 - \delta$, it achieves*

$$R_T = O \left(\sqrt{dT \left(\log \binom{d}{m} + \log \frac{1}{\delta} \right)} \right).$$

For example, one may take Algorithm 1.

We first discuss the state-of-the-art results for adversarial m -set bandits. Maiti et al. [28, Appendix E.6] obtain a high-probability regret bound of $\tilde{O}(d\sqrt{mT})$ for m -set bandits and leave open whether the extra $\sqrt{d}$ factor in this bound can be removed with an efficient algorithm. Although this consequence is not stated explicitly in their paper, applying their general DAG guarantee [28, Theorem 7] to the m -set DAG constructed in Maiti et al. [28, Appendix E.6], which has $O(md)$ edges and $K = \binom{d}{m}$ paths, and using $\log K \leq m \log(ed/m)$ yields, for $m \leq d/2$, the regret bound $\tilde{O}(m\sqrt{dT})$.

Our algorithm achieves a regret upper bound of $\tilde{O}(\sqrt{dsT})$, where $s := \min\{m, d-m\}$. In particular, when $m \leq d/2$, we have $s = m$, and the bound becomes $\tilde{O}(\sqrt{dmT})$. Thus, our bound improves the $\tilde{O}(d\sqrt{mT})$ guarantee of Maiti et al. [28, Appendix E.6] by a factor of $\sqrt{d}$, and the derived $\tilde{O}(m\sqrt{dT})$ consequence above [28, Theorem 7 and Appendix E.6] by a factor of $\sqrt{m}$. Consequently, relative to the better of these two bounds in the regime $m \leq d/2$, the improvement is a factor of $\sqrt{m}$.

²More generally, if the action losses are bounded in absolute value by L , rescaling gives the same result with the regret bound multiplied by L .

Zimmert and Lattimore [31, Theorem 6 and Algorithm 3] obtain a high-probability regret bound that, when $m \leq d/2$, yields the rate $\tilde{O}(\sqrt{dmT})$, matching our regret bound. However, their finite-action exponential-weights algorithm is stated with one weight for each of the $K = \binom{d}{m}$ actions, which can be exponential in d . Our algorithm achieves the same regret rate using only d parameters and running in polynomial time (see Section 5.3 for the implementation details).

In Table 1, we summarize the best known high-probability guarantees with $\sqrt{T}$ dependence for adversarial m -set bandits. All results reported there consider action-normalized losses and adaptive non-anticipating adversaries.

Our second contribution is a matching high-probability lower bound for the same problem.

Theorem (Informal version of Theorem 2). *There exists a universal constant $\delta_0 > 0$ such that the following holds. For every $\delta \in (0, \delta_0]$ and every horizon T above a threshold bounded by a polynomial in d , m , and $\log(1/\delta)$, every randomized m -set policy admits a deterministic adaptive non-anticipating adversary with action losses in $[0, 1]$ for which, with probability at least δ ,*

$$R_T = \Omega \left(\sqrt{dT \left(\log \binom{d}{m} + \log \frac{1}{\delta} \right)} \right).$$

Together with Theorem 1, this identifies the minimax high-probability regret up to universal constants for every d and m in the stated confidence and large-horizon regime. The special case $m = 1$ closes the remaining $\sqrt{\log K}$ gap between the confidence-tuned upper and lower bounds for ordinary adversarial bandits against adaptive non-anticipating adversaries.

Work	Regret bound	Computational complexity
Zimmert and Lattimore [31] (Theorem 6 and Algorithm 3)	$\tilde{O}(\sqrt{dmT})$	Exponential
Maiti et al. [28] (Appendix E.6)	$\tilde{O}(d\sqrt{mT})$	Polynomial
Maiti et al. [28] (Theorem 7 applied here to the DAG of Appendix E.6)	$\tilde{O}(m\sqrt{dT})$	Polynomial
This work	$\tilde{O}(\sqrt{dmT})$	Polynomial

Table 1: Statistical and computational comparison for the regime $m \leq d/2$. All upper bounds hold with high probability against adaptive non-anticipating adversaries.

Challenges and techniques. In their finite-action EXP3-KW algorithm, Zimmert and Lattimore [31, Algorithm 3] obtain a high-probability regret bound using the biased surrogate loss

$$\langle x, \hat{\ell}_t \rangle - \eta x^\top M_t^{-1} x,$$

where $\eta > 0$ is the learning rate,

$$M_t = \mathbb{E}_{X \sim p_t} [XX^\top]$$

is the second-moment matrix of the sampling distribution p_t , and

$$\hat{\ell}_t := M_t^{-1} X_t \langle X_t, \ell_t \rangle$$

is the loss estimate constructed after drawing $X_t \sim p_t$ and observing the scalar loss $\langle X_t, \ell_t \rangle$.

The main obstacle to an efficient implementation of this approach is that the leverage score is quadratic in x . Even if we start from a weighted m -set distribution and we insert this term directly into an action-level multiplicative-weights update, the updated distribution need not remain in the weighted m -set family and, consequently, it may require one weight for each of the $K = \binom{d}{m}$ actions.

Our first step is to replace the leverage score with an affine upper bound. Starting from the covariance inequality of Cesari and Colomboni [11, Corollary 1.3], we derive

$$x^\top M_t^{-1} x \leq \phi_t(x) := 1 + 2 \sum_{i=1}^d \frac{(x_i - \mu_{t,i})^2}{\mu_{t,i}(1 - \mu_{t,i})}. \quad (3)$$

Although the expression defining ϕ_t appears quadratic, it is affine on $\mathcal{X}$. Indeed, every $x \in \mathcal{X}$ satisfies $x_i^2 = x_i$ and $\sum_{i=1}^d x_i = m$. For later convenience, we scale this affine majorant by four and write

$$c_t^\top x = 4\phi_t(x) \quad \text{for every } x \in \mathcal{X}.$$

We then define the surrogate loss

$$Z_t(x) := \langle x, \hat{\ell}_t \rangle - \eta c_t^\top x. \quad (4)$$

The correction in Equation (4) plays the same role as the quadratic correction used by Zimmert and Lattimore [31, Algorithm 3], but remains affine in x . Consequently, the multiplicative-weights step maps a weighted m -set distribution to another weighted m -set distribution.

A further difficulty is that the affine leverage majorant can become arbitrarily large when a marginal probability approaches 0 or 1. Consequently, the estimated action losses $\langle x, \hat{\ell}_t \rangle$ can have arbitrarily large one-step magnitude and conditional second moment, preventing both the second-order exponential-weights analysis and the required high-probability concentration arguments. To obtain the uniform leverage bound needed in the analysis, we project the updates toward the set $\mathcal{P}_\lambda$ introduced in Section 2.

More precisely, the exact constrained OMD iterate is the KL projection of the unprojected exponential-weights distribution $\tilde{p}_{t+1}$ onto $\mathcal{P}_\lambda$. Our polynomial-time approximation is instead guaranteed to belong to $\mathcal{P}_{\lambda/2}$, where the extra slack serves to accommodate the finite accuracy of the projection procedure. The approximate KL update also preserves the weighted m -set family, as shown in Lemma 11. Hence, every iterate can be represented as p_{θ_t} using the d parameters $\theta_{t,1}, \dots, \theta_{t,d}$, rather than one weight for each of the K actions.

The comparator in the OMD analysis must also belong to $\mathcal{P}_\lambda$. However, the distribution δ_x , which puts all its probability on x , has marginal probabilities in $\{0, 1\}$ and does not belong to $\mathcal{P}_\lambda$. We therefore use the smoothed comparator

$$q_x := (1 - \lambda)\delta_x + \lambda U, \quad (5)$$

where U is the uniform distribution over $\mathcal{X}$. By construction, $q_x \in \mathcal{P}_\lambda$. Since the losses are action-normalized, using q_x in place of δ_x adds at most $2\lambda T$ to the regret.

The remaining analytical challenge is to control the estimation error under the smoothed comparator q_x , namely,

$$\sum_{t=1}^T \left(\mathbb{E}_{X \sim q_x} [\langle X, \hat{\ell}_t \rangle] - \mathbb{E}_{X \sim q_x} [\langle X, \ell_t \rangle] \right).$$

Applying a concentration bound to this estimation error introduces the positive term

$$\frac{\eta}{4} \sum_{t=1}^T \mathbb{E}_{X \sim q_x} [c_t^\top X].$$

Controlling this term separately would not give the desired regret bound. However, the affine correction in Equation (4) contributes

$$-\eta \sum_{t=1}^T \mathbb{E}_{X \sim q_x} [c_t^\top X]$$

to the OMD bound. Since $c_t^\top X = 4\phi_t(X) \geq 0$, this negative contribution dominates the positive concentration term. Their sum is therefore non-positive and may be dropped from the regret upper bound.

Finally, the elementary-symmetric-polynomial recurrence of Chen and Liu [12, Section 2, Method 2] allows us to compute the marginal probabilities and second-moment matrix of p_t in polynomial time. The associated conditional-Bernoulli procedure gives an exact sample from p_t without enumerating $\mathcal{X}$; see Chen and Liu [12, Section 4, Procedure 3], as well as the alternative backward-path construction in Procedure 5. This sampling rule is also the $L = \text{diag}(w)$ specialization of Kulesza and Taskar [26, Section 3.1 and Algorithm 2]. Together with the fact that both the affine update and the approximate KL update preserve the family of weighted m -set distributions, these routines establish the polynomial-time implementation in Proposition 1.

4 Related work

Optimal design and bandit linear optimization. Kiefer [23] introduced the general framework of approximate optimal design, and Kiefer and Wolfowitz [24] proved the equivalence between D - and G -optimality. For a finite action set $\mathcal{A} \subset \mathbb{R}^d$, Bubeck et al. [9, Theorem 4] combined exponential weights with John’s exploration and obtained $O(\sqrt{dT \log |\mathcal{A}|})$ expected regret under action-normalized losses. For general convex action sets, Abernethy et al. [2, Theorem 1] used self-concordant barriers to obtain an efficient expected-regret bound of $\tilde{O}(d^{3/2}\sqrt{T})$.

Early general high-probability guarantees with $\sqrt{T}$ dependence against adaptive adversaries had a larger polynomial dependence on the dimension. Bartlett et al. [6, Theorem 1] obtained $\tilde{O}(d^{3/2}\sqrt{T})$ regret. Abernethy and Rakhlin [1, Theorem 4 and Section 5.4] then gave a general high-probability reduction whose efficient instantiation requires problem-specific geometric subroutines. Later, Lee et al. [27, Theorem 3.1] gave the first general polynomial-time high-probability algorithm, with worst-case regret $\tilde{O}(d^{7/2}\sqrt{T})$. Zimmert and Lattimore [31, Theorem 4] improved this rate to $\tilde{O}(d^2\sqrt{T})$.

In the finite-action setting, Zimmert and Lattimore [31, Theorem 6 and Algorithm 3] combined EXP3 with Kiefer–Wolfowitz exploration and obtained $O(\sqrt{dT \log (|\mathcal{A}|/\delta)})$ regret with high probability. We use this regret bound as our statistical benchmark. When applied to $\mathcal{X} = \mathcal{X}_{d,m}$, however, its direct implementation maintains one weight for each of the $K = \binom{d}{m}$ actions, and may therefore require exponential space.

High-probability adversarial multi-armed bandits. For ordinary K -armed adversarial bandits, the standard confidence-tuned upper bound recorded in Gerchinovitz and Lattimore [16, Eq. (1)] is $O(\sqrt{KT \log(K/\delta)})$.

Gerchinovitz and Lattimore [16, Theorem 1] proved a lower bound of order $\sqrt{KT \log(1/\delta)}$, with probability at least $\delta/2$, even for a loss sequence fixed in advance. Thus, at constant confidence, the known upper and lower bounds differed by a factor of $\sqrt{\log K}$. The $m = 1$ specialization of our lower bound closes this gap against adaptive non-anticipating adversaries.

Expectation-based regret in combinatorial bandits. Early adversarial bandit algorithms for structured action sets were given by McMahan and Blum [29] and Awerbuch and Kleinberg [5]. Cesa-Bianchi and Lugosi [10] subsequently introduced ComBand, a general framework for adversarial combinatorial bandits.

The expectation-based statistical picture was further clarified by Audibert et al. [3, Section 4, after Theorem 5], who considered full-information, semi-bandit, and full-bandit feedback and established several lower bounds. Their results use coordinatewise-bounded item losses, $\ell_t \in [0, 1]^d$, so an action selecting m items may incur instantaneous loss as large as m . Under this normalization, they conjectured that the minimax bandit regret in expectation should scale as $m\sqrt{dT}$.

Under the same coordinatewise normalization, Cohen et al. [14, Theorems 1 and 5] disproved this conjecture. Using strongly correlated losses, they proved an expected-regret lower bound of

$$\Omega\left(\sqrt{\frac{dm^3T}{\log T}}\right)$$

for a multitask action class in which every action selects m items. After rescaling the losses by $1/m$ to match our action-level normalization, this lower bound becomes

$$\Omega\left(\sqrt{\frac{dmT}{\log T}}\right).$$

Ito et al. [21, Theorem 2] later removed the $1/\sqrt{\log T}$ factor using i.i.d. binary correlated losses. Crucially, their multiple-play action class is exactly $\{x \in \{0, 1\}^d : \|x\|_1 = m\}$, the m -set action class considered here. In the regime $d = \Omega(m)$ and $T = \Omega(dm^{3/2})$, their expected-regret lower bound is $\Omega(\sqrt{dm^3T})$ under coordinatewise normalization, and hence $\Omega(\sqrt{dmT})$ after rescaling to our action-level normalization.

These results identify the polynomial expected-regret hardness of the m -set problem, but they do not provide a fixed-confidence tail characterization. Indeed, after fixing a randomized hard instance by averaging, an expected lower bound $\mathbb{E}[R_T] \geq r_T$ together with the deterministic bound $R_T = O(T)$ certifies, by boundedness alone, only

$$\mathbb{P}\left(R_T \geq \frac{r_T}{2}\right) = \Omega\left(\frac{r_T}{T}\right).$$

At the rate of Ito et al. [21], this yields only a probability of order $\sqrt{dm/T}$, which may vanish as the horizon grows. In contrast, for every sufficiently small prescribed confidence level δ and all sufficiently large horizons, our lower bound gives

$$R_T = \Omega\left(\sqrt{dT\left(\log\left(\frac{d}{m}\right) + \log\frac{1}{\delta}\right)}\right)$$

with probability at least δ , while our efficient algorithm attains the matching upper bound with probability at least $1 - \delta$. Thus, we characterize the minimax fixed-confidence rate, rather than only its expectation.

Efficient algorithms for structured action sets. A separate line of work studies how to avoid explicitly enumerating exponentially large action sets. Combes et al. [15, Theorems 6 and 7] introduced CombEXP, which retains the regret scaling of earlier combinatorial-bandit algorithms

while using approximate projections and decompositions. Their implementation is polynomial-time when the action polytope admits a polynomial-size description by linear equalities and inequalities.

For general bandit linear optimization under expectation-based regret, Hazan and Karnin [19, Corollary 35] use volumetric spanners, while Ito et al. [22, Theorem 1] use a linear-optimization oracle. Under their respective access models, both approaches achieve $\tilde{O}(\sqrt{T})$ expected regret with polynomial dependence on the dimension.

Earlier efficient high-probability guarantees with $T^{2/3}$ dependence were obtained by Braun and Pokutta [8, Theorem 3.1], using a linear-optimization oracle over the action polytope, and by Sakaue et al. [30, Theorem 1], using a zero-suppressed binary decision diagram. These approaches are efficient when the required oracle or compact representation is available, but their high-probability guarantees do not attain $\sqrt{T}$ dependence.

Another recent approach is the kernelized payoff-based framework of Kontogiannis et al. [25]. Specializing their general high-probability guarantee [25, Theorem 3.2] to m -sets, for which [25, Appendix G.5] provides an efficient implementation, gives $\tilde{O}(d^{2/3}m^{4/3}T^{2/3})$ under coordinatewise-bounded item losses. After rescaling by $1/m$ to our action-level normalization, this becomes $\tilde{O}(d^{2/3}m^{1/3}T^{2/3})$. In either normalization, their horizon dependence is $T^{2/3}$, whereas ours is $\sqrt{T}$.

The work most closely related to ours is Maiti et al. [28, Appendix E.6]. Under action-normalized losses, they represent m -sets as paths in a DAG and obtain a polynomial-time high-probability algorithm against adaptive adversaries with regret $\tilde{O}(d\sqrt{mT})$. Moreover, applying their general DAG guarantee [28, Theorem 7] to the m -set DAG of [28, Appendix E.6] yields $\tilde{O}(m\sqrt{dT})$ for $m \leq d/2$. They ask whether the finite-action rate $\tilde{O}(\sqrt{dmT})$ can be achieved efficiently. Our algorithm answers this question affirmatively: it matches the finite-action EXP3-KW rate while running in polynomial time without enumerating the action set.

Weighted m -set distributions. The weighted m -set distribution introduced in Section 2 is also known as a conditional Bernoulli or rejective-sampling distribution [12, 13, 18]. Classical methods based on elementary symmetric polynomials allow us to compute its normalizing constant and marginal probabilities, and to sample from it efficiently [12, 26].

Beyond these computational properties, weighted m -set distributions have a covariance structure central to our analysis: Cesari and Colomboni [11, Corollary 1.3] prove the covariance inequality with constant $1/2$ used below.

5 An efficient minimax-optimal algorithm for m -set bandits

5.1 Affine upper bound on the leverage score

In this section, we derive the affine upper bound on the leverage score $x^\top M^{-1}x$ used by our algorithm. Complete proofs of the consequences derived here are in Section A. The covariance bound itself is imported from Cesari and Colomboni [11, Corollary 1.3] and is not reproved here.

Lemma 1 (Cesari–Colomboni covariance bound). *Let $p \in \Delta(\mathcal{X})$ be a weighted m -set distribution with positive weights, where $1 \leq m \leq d-1$, and let Σ be its covariance matrix. For every $i \in [d]$, define $v_i := \Sigma_{ii}$. Then*

$$\Sigma \succeq \frac{1}{2} \left(D - \frac{vv^\top}{V} \right),$$

where $v := (v_1, \dots, v_d)^\top$, $D := \text{diag}(v)$, and $V := \sum_{i=1}^d v_i$.

Using Lemma 1, we prove the following bound on the Moore–Penrose pseudoinverse Σ^+ of Σ .

Lemma 2. Let $p \in \Delta(\mathcal{X})$ be a weighted m -set distribution with positive weights, where $1 \leq m \leq d-1$. Let Σ be its covariance matrix, and define $v_i := \Sigma_{ii}$ for every $i \in [d]$. Then, $\ker(\Sigma) = \text{span}\{\mathbf{1}\}$, and, for every $y \in H_0 := \{z \in \mathbb{R}^d : \langle z, \mathbf{1} \rangle = 0\}$, we have

$$y^\top \Sigma^+ y \leq 2 \sum_{i=1}^d \frac{y_i^2}{v_i}.$$

Proof sketch. Let $P := D - vv^\top/V$. A direct calculation, given in [Section A.1](#), shows that $\ker(P) = \ker(\Sigma) = \text{span}\{\mathbf{1}\}$. Therefore, both matrices are positive definite, and hence invertible, on H_0 . On H_0 , the bound in [Lemma 1](#) gives

$$\Sigma \succeq \frac{1}{2}P.$$

Thus, on H_0 , we have

$$y^\top \Sigma^+ y \leq 2y^\top P^+ y \quad \text{for every } y \in H_0. \quad (6)$$

We now relate P^+ to D^{-1} . Since $y \in H_0$, we have $\sum_{i=1}^d y_i = 0$. Therefore,

$$PD^{-1}y = \left(D - \frac{vv^\top}{V}\right)D^{-1}y = y - \frac{v}{V} \sum_{i=1}^d y_i = y. \quad (7)$$

We also have

$$P(P^+y) = y, \quad (8)$$

because P^+ coincides with the inverse of P on H_0 . Subtracting [Equation \(8\)](#) from [Equation \(7\)](#) gives $P(D^{-1}y - P^+y) = 0$. It follows that

$$D^{-1}y - P^+y \in \ker(P) = \text{span}\{\mathbf{1}\}.$$

Since $y \in H_0$, this difference is orthogonal to y . Thus,

$$y^\top P^+ y = y^\top D^{-1}y = \sum_{i=1}^d \frac{y_i^2}{v_i}. \quad (9)$$

Combining [\(6\)](#) and [\(9\)](#) proves the result. See [Section A.1](#) for the complete proof. $\square$

The next lemma relates the pseudoinverse Σ^+ of the covariance matrix of a weighted m -set distribution to the inverse M^{-1} of its second-moment matrix.

Lemma 3. Let $p \in \Delta(\mathcal{X})$ be a weighted m -set distribution with positive weights and $1 \leq m \leq d-1$. Then its second-moment matrix M is positive definite and, for every $y \in H_0$,

$$y^\top M^{-1}y = y^\top \Sigma^+ y.$$

Consequently, for every $x \in \mathcal{X}$,

$$x^\top M^{-1}x = 1 + (x - \mu)^\top \Sigma^+ (x - \mu),$$

where $\mu = \mu(p)$ is the marginal vector of p .

Proof sketch. If $a^\top M a = 0$, full support implies $a^\top x = 0$ for every $x \in \mathcal{X}$. Comparing two actions that differ by swapping items i and j forces $a_i = a_j$. Hence $a = c\mathbf{1}$, and then $0 = a^\top x = cm$ gives $c = 0$. Thus $M \succ 0$. Hence, $M\mathbf{1} = \mathbb{E}_{X \sim p}[XX^\top \mathbf{1}] = m\mu$. In addition, since M is invertible,

$$M^{-1}\mu = \frac{\mathbf{1}}{m}. \quad (10)$$

Now fix $y \in H_0$. By Equation (10),

$$\mu^\top M^{-1}y = (M^{-1}\mu)^\top y = \frac{1}{m}\mathbf{1}^\top y = 0.$$

Therefore,

$$\Sigma M^{-1}y = (M - \mu\mu^\top)M^{-1}y = y.$$

Since Σ^+ is the inverse of Σ on H_0 , the vectors $M^{-1}y$ and Σ^+y differ by an element of $\ker(\Sigma) = \text{span}\{\mathbf{1}\}$. This difference is orthogonal to y , and thus

$$y^\top M^{-1}y = y^\top \Sigma^+y. \quad (11)$$

Finally, fix $x \in \mathcal{X}$. Since both x and μ have coordinates that sum to m , we have $x - \mu \in H_0$. Expanding $x = \mu + (x - \mu)$ and using Equation (10) and Equation (11) gives

$$x^\top M^{-1}x = 1 + (x - \mu)^\top \Sigma^+(x - \mu).$$

See Section A.2 for the complete proof. $\square$

We now combine Lemmas 2 and 3 to obtain the affine upper bound on the leverage score used by our algorithm.

Lemma 4. *Let $p \in \mathcal{P}_\lambda$ be a weighted m -set distribution, where $1 \leq m \leq d - 1$, and let μ and M be its marginal vector and second-moment matrix. For every $x \in \mathcal{X}$, define*

$$\phi_p(x) := 1 + 2 \sum_{i=1}^d \frac{(x_i - \mu_i)^2}{\mu_i(1 - \mu_i)}.$$

Then, for every $x \in \mathcal{X}$, $x^\top M^{-1}x \leq \phi_p(x)$. Moreover, $4\phi_p(x) = c_p^\top x$, where

$$c_{p,i} = 4 \left(\frac{1 + 2 \sum_{j=1}^d \frac{\mu_j}{1 - \mu_j}}{m} + 2 \frac{1 - 2\mu_i}{\mu_i(1 - \mu_i)} \right) \quad \text{for every } i \in [d].$$

Finally, $\mathbb{E}_{X \sim p}[\phi_p(X)] = 2d + 1$, and, for every $x \in \mathcal{X}$, $0 \leq \phi_p(x) \leq 1 + 4d/\lambda$.

Proof sketch. Fix $x \in \mathcal{X}$. Since both x and μ have coordinates that sum to m , we have $x - \mu \in H_0$. Therefore, Lemmas 2 and 3 give

$$x^\top M^{-1}x = 1 + (x - \mu)^\top \Sigma^+(x - \mu) \leq 1 + 2 \sum_{i=1}^d \frac{(x_i - \mu_i)^2}{\mu_i(1 - \mu_i)} = \phi_p(x).$$

Since $\mathbb{E}_{X \sim p}[(X_i - \mu_i)^2] = \mu_i(1 - \mu_i)$, we have $\mathbb{E}_{X \sim p}[\phi_p(X)] = 1 + 2d$. Moreover, $p \in \mathcal{P}_\lambda$ implies $\mu_i \geq \lambda r$ and $1 - \mu_i \geq \lambda(1 - r)$, where $r = m/d$. Separating the m coordinates for which $x_i = 1$ from the $d - m$ coordinates for which $x_i = 0$ gives

$$\sum_{i=1}^d \frac{(x_i - \mu_i)^2}{\mu_i(1 - \mu_i)} \leq \frac{m}{\lambda r} + \frac{d - m}{\lambda(1 - r)} = \frac{2d}{\lambda}.$$

Hence,

$$0 \leq \phi_p(x) \leq 1 + \frac{4d}{\lambda}.$$

Finally, using $x_i^2 = x_i$, we obtain

$$\phi_p(x) = 1 + 2 \sum_{j=1}^d \frac{\mu_j}{1 - \mu_j} + 2 \sum_{i=1}^d \frac{1 - 2\mu_i}{\mu_i(1 - \mu_i)} x_i.$$

Since $\sum_i x_i = m$, we can write the constant term as a linear function of x . Multiplying the resulting affine representation by four gives $4\phi_p(x) = c_p^\top x$, with c_p defined in the statement. See [Section A.3](#) for the complete proof. $\square$

5.2 Algorithm and high-probability regret

In this section, we present [Algorithm 1](#) and prove its high-probability regret guarantee. At each round t , the learner draws an action X_t from a weighted m -set distribution $p_t \in \mathcal{P}_{\lambda/2}$ and observes only the scalar loss $\langle X_t, \ell_t \rangle$. It uses this observation to construct the KW loss estimate $\widehat{\ell}_t$.

At round t , we set $\phi_t := \phi_{p_t}$ and $c_t := c_{p_t}$, where ϕ_p and c_p are defined in [Lemma 4](#). The learner forms a biased surrogate loss by subtracting the affine correction $\eta c_t^\top x$ from $\langle x, \widehat{\ell}_t \rangle$. Finally, the learner computes an approximate entropic OMD update toward $\mathcal{P}_\lambda$ (defined in [Eq. \(2\)](#)). The update is accurate enough for the OMD analysis and returns a distribution in $\mathcal{P}_{\lambda/2}$, so its marginal probabilities stay away from 0 and 1. We state the resulting regret bound in [Theorem 1](#).

Theorem 1 (High-probability KW bound). *There exists a universal constant $C > 0$ such that the following holds. Let $T \geq 1$ be an integer, let $\delta \in (0, 1)$, and assume $1 \leq m \leq d - 1$. Run [Algorithm 1](#). Then, with probability at least $1 - \delta$,*

$$R_T \leq C \sqrt{dT \left(\log K + \log \frac{1}{\delta} \right)},$$

where $K = \binom{d}{m}$. One may take $C = 160$.

Proof sketch. For every $x \in \mathcal{X}$, define the smoothed comparator

$$q_x := (1 - \lambda)\delta_x + \lambda U,$$

where δ_x puts all its probability on x and U is the uniform distribution over $\mathcal{X}$. By construction, $q_x \in \mathcal{P}_\lambda$. The concentration bounds below hold simultaneously for every $x \in \mathcal{X}$, so we can then choose x to be a best action.

By adding and subtracting the expected losses under p_t and q_x , and applying [Lemma 9](#), we obtain, with high probability,

$$R_T \leq \underbrace{\sum_{t=1}^T (\mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle] - \mathbb{E}_{X \sim q_x}[\langle X, \ell_t \rangle])}_{=:(\star)} + O \left(\sqrt{T \log \frac{1}{\delta}} \right) + O(\lambda T). \quad (12)$$

The square-root term controls the difference between the learner's realized and expected losses. The term $O(\lambda T)$ is the cost of using q_x in place of x . It remains to bound $(\star)$. Adding and subtracting

Algorithm 1 Affine KW-OMD with approximate entropic updates toward $\mathcal{P}_\lambda$

Require: $T \geq 1, \delta \in (0, 1)$

- 1: Set $\eta \leftarrow \min \left\{ \frac{1}{256d}, \sqrt{\frac{\log(12K/\delta)}{320dT}} \right\}$
- 2: Set $\lambda \leftarrow 128\eta d$ and $\varepsilon_p \leftarrow \eta/T$
- 3: Initialize $\theta_1 \leftarrow 0$ and $p_1 \leftarrow p_{\theta_1} = U$ $\triangleright U$ is a uniform distribution
- 4: **for** $t = 1, \dots, T$ **do**
- 5: Compute $\mu_t \leftarrow \mu(p_t)$, $M_t \leftarrow M(p_t) = \mathbb{E}_{X \sim p_t}[XX^\top]$
- 6: Set $\phi_t \leftarrow \phi_{p_t}$ and $c_t \leftarrow c_{p_t}$, so that, for every $x \in \mathcal{X}$,

$$\phi_t(x) \leftarrow 1 + 2 \sum_{i=1}^d \frac{(x_i - \mu_{t,i})^2}{\mu_{t,i}(1 - \mu_{t,i})} \quad \text{and} \quad c_t^\top x = 4\phi_t(x)$$

- 7: Draw $X_t \sim p_t$
- 8: Observe $Y_t = \langle X_t, \ell_t \rangle$
- 9: Compute $\hat{\ell}_t \leftarrow M_t^{-1} X_t Y_t$
- 10: Define $Z_t(x) \leftarrow \langle x, \hat{\ell}_t \rangle - \eta c_t^\top x$, for every $x \in \mathcal{X}$
- 11: Set $z_t \leftarrow \hat{\ell}_t - \eta c_t$, $\tilde{\theta}_{t+1} \leftarrow \theta_t - \eta z_t$, and $\tilde{p}_{t+1} \leftarrow p_{\tilde{\theta}_{t+1}}$
- 12: Apply a fixed deterministic implementation of the procedure in [Lemma 11](#) to obtain $p_{t+1} = p_{\theta_{t+1}}$ satisfying

$$p_{t+1} \in \mathcal{P}_{\lambda/2}$$

and, simultaneously for every $q \in \mathcal{P}_\lambda$,

$$\text{KL}(q \| p_{t+1}) + \text{KL}(p_{t+1} \| \tilde{p}_{t+1}) \leq \text{KL}(q \| \tilde{p}_{t+1}) + \varepsilon_p.$$

13: **end for**

the expectations of the estimated losses gives

$$\begin{aligned} (\star) &= \sum_{t=1}^T \left(\mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle] - \mathbb{E}_{X \sim p_t}[\langle X, \hat{\ell}_t \rangle] \right) \\ &\quad + \sum_{t=1}^T \left(\mathbb{E}_{X \sim p_t}[\langle X, \hat{\ell}_t \rangle] - \mathbb{E}_{X \sim q_x}[\langle X, \hat{\ell}_t \rangle] \right) + \sum_{t=1}^T \left(\mathbb{E}_{X \sim q_x}[\langle X, \hat{\ell}_t \rangle] - \mathbb{E}_{X \sim q_x}[\langle X, \ell_t \rangle] \right). \end{aligned} \quad (13)$$

The first and third terms in (13) can be bounded by applying [Lemma 7](#) under p_t and q_x , respectively. To bound the second term, recall the surrogate loss

$$Z_t(x) := \langle x, \hat{\ell}_t \rangle - \eta c_t^\top x, \quad x \in \mathcal{X}.$$

Since

$$\langle x, \hat{\ell}_t \rangle = Z_t(x) + \eta c_t^\top x,$$

we can apply [Lemma 6](#) and [Lemma 8](#) to the surrogate losses Z_t . Moreover, [Lemma 4](#) gives

$$\mathbb{E}_{X \sim p_t}[c_t^\top X] = 4(2d + 1) = O(d).$$

Combining these results, we have

$$(\star) \leq O\left(\frac{\log K + \log(1/\delta)}{\eta} + \eta dT + 1\right) + \underbrace{\left(-\eta + \frac{\eta}{4}\right) \sum_{t=1}^T \mathbb{E}_{X \sim q_x} [c_t^\top X]}_{=:(\star\star) \leq 0}. \quad (14)$$

The coefficient $-\eta$ in $(\star\star)$ comes from the affine correction $-\eta c_t^\top x$ in Z_t , while the coefficient $\eta/4$ comes from the concentration bound for the third term in (13). Since

$$c_t^\top X = 4\phi_t(X) \geq 0,$$

the term $(\star\star)$ is nonpositive. The additive constant in (14) is the accumulated error of the approximate KL updates: by the choice $\varepsilon_p = \eta/T$, its total contribution is $T\varepsilon_p/\eta = 1$. Substituting (14) into (12) gives

$$R_T \leq O\left(\frac{\log K + \log(1/\delta)}{\eta} + \eta dT + 1 + \lambda T + \sqrt{T \log \frac{1}{\delta}}\right). \quad (15)$$

Finally, substituting $\lambda = 128\eta d$ and the value of η , and combining the resulting high-probability estimate with the deterministic bound $R_T \leq 2T$, gives the stated result with $C = 160$. The complete two-case calculation is given in Section B.1. $\square$

Let $s := \min\{m, d - m\}$. Since $\log K \leq s \log(ed/s)$, the bound in Theorem 1 is $\tilde{O}(\sqrt{dsT})$. In particular, when $m \leq d/2$, it is $\tilde{O}(\sqrt{dmT})$.

5.3 Polynomial-time implementation

We now show that Algorithm 1 can be implemented in polynomial time without enumerating $\mathcal{X}$.

The key idea is to maintain p_t in the weighted m -set form p_{θ_t} , with $\theta_t \in \mathbb{R}^d$, thereby allowing all steps of the algorithm to be performed without enumerating $\mathcal{X}$. The convex problem used in the update is solved to the explicit inverse-polynomial objective accuracy established in Lemma 11.

Proposition 1. *Algorithm 1 can be implemented in $T \cdot \text{poly}(d, m, \log T)$ time and $\text{poly}(d, m)$ space, without enumerating $\mathcal{X}$. At every round, p_t is a weighted m -set distribution p_{θ_t} represented by the d parameters $\theta_{t,1}, \dots, \theta_{t,d}$.*

Proof sketch. We observe that if p_t has a weighted m -set representation, its moments can be computed and a sample can be drawn in polynomial time using the recurrences in Lemma 12.

The strategy update in Algorithm 1 can be decomposed into two steps. First, define the exponential update

$$\tilde{p}_{t+1}(x) \propto p_t(x) \exp(-\eta Z_t(x)).$$

The exact update toward which we compute is the KL projection

$$p_{t+1}^* \in \arg \min_{p \in \mathcal{P}_\lambda} \text{KL}(p \| \tilde{p}_{t+1}).$$

Since Z_t is affine, the exponential update only changes the d parameters, and $\tilde{p}_{t+1}$ remains a weighted m -set distribution. The exact projection is characterized by a convex problem in $2d$ variables. Solving this problem to the objective accuracy in Lemma 11 returns a weighted m -set distribution $p_{t+1} \in \mathcal{P}_{\lambda/2}$ and introduces at most $\varepsilon_p = \eta/T$ error in the KL inequality used by OMD. The required objective accuracy is inverse-polynomial, and the ellipsoid method obtains it using

polynomially many evaluations of the objective and its gradient. Since the initial distribution p_1 is uniform and admits a weighted m -set representation, this representation is preserved at every round. Thus, all steps of the algorithm can be performed in polynomial time without enumerating $\mathcal{X}$. The complete proof is given in [Section C.3](#). $\square$

5.4 Matching high-probability lower bound

We complement the upper bound with a matching high-probability lower bound.

Theorem 2. *There exist universal constants $c > 0$ and $\delta_0 \in (0, 1)$ and a threshold $\mathfrak{T}(d, m, \delta)$ bounded by a polynomial in d , m , and $\log(1/\delta)$ such that the following holds. Let $d \geq 2$, $1 \leq m \leq d - 1$, $\delta \in (0, \delta_0]$, and $T \geq \mathfrak{T}(d, m, \delta)$. For every randomized policy on $\mathcal{X}_{d,m}$, there exists a deterministic adaptive non-anticipating loss process whose action losses belong to $[0, 1]$ and such that*

$$\mathbb{P} \left(R_T \geq c \sqrt{dT \left(\log \binom{d}{m} + \log \frac{1}{\delta} \right)} \right) \geq \delta.$$

Proof sketch. Let $s := \min\{m, d - m\} \leq d/2$. By complementation, it is enough to prove the result for s -set actions. We establish separately the contributions depending on the number of actions and on the confidence level.

For the action-count term, suppose first that d/s is large. We divide the items into s groups and construct the losses so that one item in each group is slightly better than the others. The item losses share a common random component, which can be adjusted when the favorable item is changed so that the learner observes the same aggregate feedback. Thus, the learner cannot determine which item is favorable in each group, yielding the $\log \binom{d}{m}$ term. When d/s is small, we instead arrange the items into pairs and make one item in each pair slightly better. The aggregate feedback does not reveal enough information to identify all these pairwise choices, which gives the same action-count term.

For the confidence term, we fix $s - 1$ core items. Selecting multiple items outside the core produces an immediate additional loss. Otherwise, the learner can test only one candidate per round, so some candidate is selected in only a small fraction of the rounds. We make this candidate favorable. Since the learner observes different feedback only when it selects the candidate, the favorable environment remains difficult to distinguish from the baseline one, yielding the full $\log(1/\delta)$ dependence.

We use the action-count or confidence construction depending on which logarithmic term is larger. This gives the claimed lower bound up to a universal constant and requires only a polynomial lower threshold on T . All action losses belong to $[0, 1]$, and complementation preserves the learner's feedback and regret. The complete proof is given in the appendix. $\square$

Together, [Theorems 1](#) and [2](#) show that for every d and m , for all sufficiently small δ , and for horizons above a polynomial threshold, the minimax high-probability regret is

$$\Theta \left(\sqrt{dT \left(\log \binom{d}{m} + \log \frac{1}{\delta} \right)} \right).$$

For a strict comparison of the two tail probabilities, apply the lower bound with confidence level 2δ . This changes $\log(1/\delta)$ only by the additive constant $\log 2$.

When $m = 1$ and $d = K$, our problem reduces to the standard adversarial K -armed bandit problem.

Corollary 1. *There exist universal constants $c > 0$ and $\delta_0 \in (0, 1)$ and a threshold $\mathfrak{T}_{\text{MAB}}(K, \delta)$ bounded by a polynomial in K and $\log(1/\delta)$ such that the following holds. Let $K \geq 2$, $\delta \in (0, \delta_0]$, and $T \geq \mathfrak{T}_{\text{MAB}}(K, \delta)$. For every randomized K -armed bandit policy, there exists a deterministic adaptive non-anticipating loss process $\ell_t \in [0, 1]^K$ such that*

$$\mathbb{P} \left(R_T \geq c \sqrt{KT \left(\log K + \log \frac{1}{\delta} \right)} \right) \geq \delta.$$

Proof. Apply [Theorem 2](#) with $d = K$ and $m = 1$, and set $\mathfrak{T}_{\text{MAB}}(K, \delta) := \mathfrak{T}(K, 1, \delta)$. $\square$

When $d = K$ and $m = 1$, our protocol coincides with the standard K -armed bandit protocol considered by Gerchinovitz and Lattimore [16], in which the adversary may observe the learner’s distribution before choosing the loss vector. Indeed, in both models, the learner first selects a distribution p_t , after which the adversary chooses ℓ_t , possibly as a function of p_t , but before the learner draws an arm from p_t .

Equation (1) of Gerchinovitz and Lattimore [16] states that, if δ is fixed in advance and used to set the parameters of the algorithm, then, with probability at least $1 - \delta$, the regret is at most $O(\sqrt{KT \log(K/\delta)})$. Their Theorem 1 instead shows that, even for a loss sequence fixed in advance, every randomized algorithm can suffer regret $\Omega(\sqrt{KT \log(1/\delta)})$ with probability of order δ . For fixed δ , these bounds differ by a factor of $\sqrt{\log K}$.

Our corollary closes this gap when the adversary is adaptive and non-anticipating. For every randomized K -armed bandit policy and all sufficiently large horizons, it constructs such an adversary for which

$$\mathbb{P} \left(R_T \geq c \sqrt{KT \left(\log K + \log \frac{1}{\delta} \right)} \right) \geq \delta.$$

This matches the existing upper bound up to universal constants.

6 Conclusion and future work

We introduced an algorithm for adversarial m -set bandits whose high-probability regret matches the finite-action EXP3-KW rate up to logarithmic factors, while running in polynomial time without enumerating the action set. We complemented this upper bound with a matching lower bound showing that

$$\sqrt{dT \left(\log \binom{d}{m} + \log \frac{1}{\delta} \right)}$$

is the minimax high-probability rate up to universal constants for every d and m , for all sufficiently small δ , and for horizons above a polynomial threshold.

The special case $m = 1$ closes the remaining gap between the known upper and lower bounds for ordinary adversarial bandits against adaptive non-anticipating adversaries. More precisely, Gerchinovitz and Lattimore [16, Eq. (1) and Theorem 1] left a $\sqrt{\log K}$ gap at constant confidence between the standard high-probability upper bound and their lower bound. Our lower bound shows that, the $\log K$ factor is necessary for fixed sufficiently small confidence levels and sufficiently large horizons.

As further directions, it would be interesting to determine whether our approach extends beyond m -sets. A natural test case is that of matroid-base action sets, since m -sets are precisely the bases

of a uniform matroid. Although affine exponential-weights updates still preserve weighted matroid-base distributions, our analysis also requires an efficiently computable affine leverage majorant with controlled expectation and range, as well as efficient moment and KL-projection routines. These properties do not follow from our present covariance-based construction, so extending the near-optimal high-probability guarantee to broader matroid classes would require additional structural and computational ideas.

Appendix

A Proofs from Section 5.1

A.1 Proof of Lemma 2

Lemma 2. *Let $p \in \Delta(\mathcal{X})$ be a weighted m -set distribution with positive weights, where $1 \leq m \leq d-1$. Let Σ be its covariance matrix, and define $v_i := \Sigma_{ii}$ for every $i \in [d]$. Then, $\ker(\Sigma) = \text{span}\{\mathbf{1}\}$, and, for every $y \in H_0 := \{z \in \mathbb{R}^d : \langle z, \mathbf{1} \rangle = 0\}$, we have*

$$y^\top \Sigma^+ y \leq 2 \sum_{i=1}^d \frac{y_i^2}{v_i}.$$

Proof. Let

$$P := D - \frac{vv^\top}{V}.$$

We first identify the kernels of P and Σ . Since p has positive weights, it has full support on $\mathcal{X}$. Therefore,

$$z^\top \Sigma z = \text{Var}_{X \sim p}(\langle z, X \rangle) = 0$$

only if $\langle z, x \rangle$ is the same for every $x \in \mathcal{X}$.

Fix $i \neq j$. Since $1 \leq m \leq d-1$, we can choose two actions that differ only by exchanging item i with item j . The equality of their inner products with z gives $z_i = z_j$. Thus, all the coordinates of z are equal. Conversely, $\langle \mathbf{1}, X \rangle = m$ for every $X \in \mathcal{X}$, so $\Sigma \mathbf{1} = 0$. Hence,

$$\ker(\Sigma) = \text{span}\{\mathbf{1}\}. \quad (16)$$

Moreover, for every $z \in \mathbb{R}^d$,

$$z^\top P z = \sum_{i=1}^d v_i z_i^2 - \frac{\left(\sum_{i=1}^d v_i z_i\right)^2}{V} = \sum_{i=1}^d v_i (z_i - \bar{z}_v)^2,$$

where

$$\bar{z}_v := \frac{\sum_{i=1}^d v_i z_i}{V}.$$

Full support also gives

$$v_i = \text{Var}(X_i) = \mu_i(1 - \mu_i) > 0 \quad \text{for every } i \in [d].$$

It follows that $z^\top P z = 0$ if and only if all the coordinates of z are equal. Therefore,

$$\ker(P) = \text{span}\{\mathbf{1}\}. \quad (17)$$

By (16) and (17), the pseudoinverse order property stated in the preliminaries applies to the bound in Lemma 1. Hence, for every $y \in H_0$,

$$y^\top \Sigma^+ y \leq 2y^\top P^+ y. \quad (18)$$

It remains to compute $y^\top P^+ y$. Since $D = \text{diag}(v)$, we have $v^\top D^{-1} = \mathbf{1}^\top$. Thus, for every $y \in H_0$,

$$PD^{-1}y = y - \frac{v}{V} \mathbf{1}^\top y = y. \quad (19)$$

Moreover, $\text{range}(P) = H_0$, so

$$PP^+y = y. \quad (20)$$

Equations (19) and (20) show that $D^{-1}y$ and P^+y both solve the equation $Pz = y$. Therefore,

$$D^{-1}y - P^+y \in \ker(P) = \text{span}\{\mathbf{1}\}.$$

Since $y \in H_0$, it is orthogonal to this difference. Consequently,

$$y^\top P^+y = y^\top D^{-1}y = \sum_{i=1}^d \frac{y_i^2}{v_i}. \quad (21)$$

Combining (18) and (21) proves the result. $\square$

A.2 Proof of Lemma 3

Lemma 3. *Let $p \in \Delta(\mathcal{X})$ be a weighted m -set distribution with positive weights and $1 \leq m \leq d-1$. Then its second-moment matrix M is positive definite and, for every $y \in H_0$,*

$$y^\top M^{-1}y = y^\top \Sigma^+y.$$

Consequently, for every $x \in \mathcal{X}$,

$$x^\top M^{-1}x = 1 + (x - \mu)^\top \Sigma^+(x - \mu),$$

where $\mu = \mu(p)$ is the marginal vector of p .

Proof. We first show that M is positive definite. Suppose that $a^\top Ma = 0$. Since

$$a^\top Ma = \mathbb{E}_{X \sim p}[(a^\top X)^2],$$

we have $a^\top x = 0$ for every $x \in \mathcal{X}$, because p has full support.

Fix $i \neq j$. Since $1 \leq m \leq d-1$, we can choose two actions that differ only by exchanging item i with item j . Comparing their inner products with a gives $a_i = a_j$. Thus, $a = c\mathbf{1}$ for some $c \in \mathbb{R}$. Since every $x \in \mathcal{X}$ has m nonzero coordinates,

$$0 = a^\top x = cm.$$

Therefore, $c = 0$, and hence $a = 0$. This proves that $M \succ 0$. By Lemma 2, $\ker(\Sigma) = \text{span}\{\mathbf{1}\}$. Since Σ is symmetric, this also gives $\text{range}(\Sigma) = H_0$. Every $X \in \mathcal{X}$ satisfies $\langle X, \mathbf{1} \rangle = m$. Therefore,

$$M\mathbf{1} = \mathbb{E}_{X \sim p}[X\langle X, \mathbf{1} \rangle] = m\mu.$$

Since M is invertible,

$$M^{-1}\mu = \frac{1}{m}\mathbf{1}. \quad (22)$$

Now fix $y \in H_0$ and let $z := M^{-1}y$. By (22),

$$\mu^\top z = (M^{-1}\mu)^\top y = \frac{1}{m}\mathbf{1}^\top y = 0.$$

It follows that

$$\Sigma z = (M - \mu\mu^\top)z = y.$$

Moreover, $\Sigma^+ y$ also solves the equation $\Sigma u = y$. Therefore,

$$z - \Sigma^+ y \in \ker(\Sigma) = \text{span}\{\mathbf{1}\}.$$

Since $y \in H_0$, it is orthogonal to this difference. Hence,

$$y^\top M^{-1} y = y^\top \Sigma^+ y. \quad (23)$$

Finally, fix $x \in \mathcal{X}$ and let $y := x - \mu$. Since $\mathbf{1}^\top x = \mathbf{1}^\top \mu = m$, we have $y \in H_0$. Expanding $x = \mu + y$ gives

$$\begin{aligned} x^\top M^{-1} x &= \mu^\top M^{-1} \mu + 2y^\top M^{-1} \mu + y^\top M^{-1} y \\ &= 1 + y^\top \Sigma^+ y, \end{aligned}$$

where the second equality follows from (22) and (23). Substituting $y = x - \mu$ proves that

$$x^\top M^{-1} x = 1 + (x - \mu)^\top \Sigma^+ (x - \mu).$$

This concludes the proof. $\square$

A.3 Proof of Lemma 4

Lemma 4. *Let $p \in \mathcal{P}_\lambda$ be a weighted m -set distribution, where $1 \leq m \leq d-1$, and let μ and M be its marginal vector and second-moment matrix. For every $x \in \mathcal{X}$, define*

$$\phi_p(x) := 1 + 2 \sum_{i=1}^d \frac{(x_i - \mu_i)^2}{\mu_i(1 - \mu_i)}.$$

Then, for every $x \in \mathcal{X}$, $x^\top M^{-1} x \leq \phi_p(x)$. Moreover, $4\phi_p(x) = c_p^\top x$, where

$$c_{p,i} = 4 \left(\frac{1 + 2 \sum_{j=1}^d \frac{\mu_j}{1 - \mu_j}}{m} + 2 \frac{1 - 2\mu_i}{\mu_i(1 - \mu_i)} \right) \quad \text{for every } i \in [d].$$

Finally, $\mathbb{E}_{X \sim p}[\phi_p(X)] = 2d + 1$, and, for every $x \in \mathcal{X}$, $0 \leq \phi_p(x) \leq 1 + 4d/\lambda$.

Proof. For $x \in \mathcal{X}$, put $y = x - \mu \in H_0$. By Lemma 3,

$$x^\top M^{-1} x = 1 + y^\top \Sigma^+ y.$$

Lemma 2 gives

$$y^\top \Sigma^+ y \leq 2 \sum_i \frac{y_i^2}{\mu_i(1 - \mu_i)},$$

and hence $x^\top M^{-1} x \leq \phi_p(x)$. Taking expectations and using $\mathbb{E}[(X_i - \mu_i)^2] = \mu_i(1 - \mu_i)$ yields

$$\mathbb{E}_{X \sim p}[\phi_p(X)] = 2d + 1.$$

For the range bound, if $x_i = 1$, then

$$2 \frac{(x_i - \mu_i)^2}{\mu_i(1 - \mu_i)} = 2 \frac{1 - \mu_i}{\mu_i} \leq \frac{2}{\lambda r};$$

if $x_i = 0$, then

$$2 \frac{(x_i - \mu_i)^2}{\mu_i(1 - \mu_i)} = 2 \frac{\mu_i}{1 - \mu_i} \leq \frac{2}{\lambda(1 - r)}.$$

Since x has m ones and $d - m$ zeros,

$$\phi_p(x) \leq 1 + \frac{2m}{\lambda r} + \frac{2(d - m)}{\lambda(1 - r)} = 1 + \frac{4d}{\lambda}.$$

Finally, because $x_i \in \{0, 1\}$,

$$2 \frac{(x_i - \mu_i)^2}{\mu_i(1 - \mu_i)} = 2 \frac{\mu_i}{1 - \mu_i} + 2x_i \frac{1 - 2\mu_i}{\mu_i(1 - \mu_i)}.$$

Thus

$$\phi_p(x) = 1 + 2 \sum_i \frac{\mu_i}{1 - \mu_i} + 2 \sum_i x_i \frac{1 - 2\mu_i}{\mu_i(1 - \mu_i)}.$$

Distribute the constant term evenly over the selected coordinates using $\sum_i x_i = m$, and multiply the resulting affine representation by four, to obtain $4\phi_p(x) = c_p^\top x$. $\square$

B Proofs from Section 5.2

Assumption 1. Throughout this appendix, $T \geq 1$ is an integer, $\delta \in (0, 1)$,

$$0 < \eta \leq \frac{1}{64}, \quad \lambda = 128\eta d \leq \frac{1}{2}.$$

The loss sequence $(\ell_t)_{t=1}^T$ is predictable, that is, ℓ_t is $\mathcal{F}_{t-1}$ -measurable for every $t \in [T]$, and satisfies the action-normalized condition in [Section 2](#).

The initialization and the update in [Algorithm 1](#) ensure that, at every round t , the sampling distribution p_t is a weighted m -set distribution with positive weights and $p_t \in \mathcal{P}_{\lambda/2}$. Hence, the assumptions of [Lemmas 1](#) and [4](#) hold at every round.

Recall from [Algorithm 1](#) that

$$\phi_t := \phi_{p_t}, \quad c_t := c_{p_t}, \quad c_t^\top x = 4\phi_t(x), \quad Z_t(x) := \hat{y}_t(x) - \eta c_t^\top x.$$

We also write

$$\hat{y}_t(x) := \langle x, \hat{\ell}_t \rangle = x^\top M_t^{-1} X_t Y_t.$$

Set

$$B := 1 + \frac{8d}{\lambda}.$$

Applying [Lemma 4](#) with $\lambda/2$, for every $x \in \mathcal{X}$, we have

$$0 \leq c_t^\top x \leq 4B, \quad \mathbb{E}_{X \sim p_t}[c_t^\top X] = 4(2d + 1), \quad x^\top M_t^{-1} x \leq \frac{c_t^\top x}{4}.$$

Lemma 5. Fix $t \in [T]$. Suppose that $0 < \eta \leq 1/64$, $\lambda = 128\eta d \leq 1/2$, the loss at round t satisfies the action-normalized condition, and p_t is a weighted m -set distribution with positive weights in $\mathcal{P}_{\lambda/2}$. Then, for every $x \in \mathcal{X}$, $\eta|Z_t(x)| \leq 1/4$.

Proof. Since $|Y_t| \leq 1$, the Cauchy-Schwarz inequality in the norm induced by M_t^{-1} gives

$$|\hat{y}_t(x)| = |x^\top M_t^{-1} X_t Y_t| \leq \sqrt{x^\top M_t^{-1} x} \sqrt{X_t^\top M_t^{-1} X_t}.$$

Applying [Lemma 4](#) to $p_t \in \mathcal{P}_{\lambda/2}$, both leverage scores in the last inequality are at most $\phi_t(x)$ and $\phi_t(X_t)$, respectively, and both ϕ_t -values are at most

$$B = 1 + \frac{8d}{\lambda}.$$

Thus

$$|\hat{y}_t(x)| \leq B.$$

Also $0 \leq c_t^\top x \leq 4B$. Hence

$$|Z_t(x)| = |\hat{y}_t(x) - \eta c_t^\top x| \leq B + 4\eta B.$$

The parameter choice implies

$$\eta B = \eta + \frac{8\eta d}{\lambda} = \eta + \frac{1}{16} \leq \frac{5}{64}.$$

Therefore

$$\eta |Z_t(x)| \leq (1 + 4\eta)\eta B \leq \left(1 + \frac{1}{16}\right) \frac{5}{64} = \frac{85}{1024} < \frac{1}{4},$$

where we used $\eta \leq 1/64$. This proves the claim. $\square$

Lemma 6. Under [Assumption 1](#), for every distribution $q \in \mathcal{P}_\lambda$,

$$\sum_{t=1}^T (\mathbb{E}_{X \sim p_t}[Z_t(X)] - \mathbb{E}_{X \sim q}[Z_t(X)]) \leq \frac{\text{KL}(q \| p_1)}{\eta} + 1 + 2\eta \sum_{t=1}^T \mathbb{E}_{X \sim p_t}[Z_t(X)^2].$$

Proof. Fix $t \in [T]$ and define the unprojected multiplicative-weights distribution

$$\tilde{p}_{t+1}(x) := \frac{p_t(x) \exp(-\eta Z_t(x))}{\mathbb{E}_{X \sim p_t}[\exp(-\eta Z_t(X))]}, \quad x \in \mathcal{X}.$$

Since $Z_t(x) = z_t^\top x$ and $p_t = p_{\theta_t}$, the distribution above is precisely $p_{\tilde{\theta}_{t+1}}$ from [Algorithm 1](#). The guarantee in [Algorithm 1](#) therefore gives, for every $q \in \mathcal{P}_\lambda$,

$$\text{KL}(q \| p_{t+1}) \leq \text{KL}(q \| \tilde{p}_{t+1}) + \varepsilon_p, \tag{24}$$

where we dropped the nonnegative term $\text{KL}(p_{t+1} \| \tilde{p}_{t+1})$. By the definition of $\tilde{p}_{t+1}$,

$$\text{KL}(q \| \tilde{p}_{t+1}) = \text{KL}(q \| p_t) + \eta \mathbb{E}_{X \sim q}[Z_t(X)] + \log \mathbb{E}_{X \sim p_t}[\exp(-\eta Z_t(X))].$$

Combining this identity with (24) and rearranging gives

$$\begin{aligned} \eta (\mathbb{E}_{X \sim p_t}[Z_t(X)] - \mathbb{E}_{X \sim q}[Z_t(X)]) &\leq \text{KL}(q \| p_t) - \text{KL}(q \| p_{t+1}) \\ &\quad + \eta \mathbb{E}_{X \sim p_t}[Z_t(X)] + \log \mathbb{E}_{X \sim p_t}[\exp(-\eta Z_t(X))] + \varepsilon_p. \end{aligned}$$

By [Lemma 5](#),

$$|\eta Z_t(x)| \leq \frac{1}{4} \quad \text{for every } x \in \mathcal{X}.$$

Using

$$e^{-u} \leq 1 - u + 2u^2 \quad \text{for } |u| \leq \frac{1}{4},$$

we obtain

$$\mathbb{E}_{X \sim p_t}[\exp(-\eta Z_t(X))] \leq 1 - \eta \mathbb{E}_{X \sim p_t}[Z_t(X)] + 2\eta^2 \mathbb{E}_{X \sim p_t}[Z_t(X)^2].$$

The inequality $\log(1 + s) \leq s$ then gives

$$\eta \mathbb{E}_{X \sim p_t}[Z_t(X)] + \log \mathbb{E}_{X \sim p_t}[\exp(-\eta Z_t(X))] \leq 2\eta^2 \mathbb{E}_{X \sim p_t}[Z_t(X)^2].$$

Therefore,

$$\mathbb{E}_{X \sim p_t}[Z_t(X)] - \mathbb{E}_{X \sim q}[Z_t(X)] \leq \frac{\text{KL}(q \| p_t) - \text{KL}(q \| p_{t+1})}{\eta} + 2\eta \mathbb{E}_{X \sim p_t}[Z_t(X)^2] + \frac{\varepsilon_p}{\eta}.$$

Summing over t telescopes the KL terms. Finally, $\text{KL}(q \| p_{T+1}) \geq 0$, so

$$\sum_{t=1}^T (\mathbb{E}_{X \sim p_t}[Z_t(X)] - \mathbb{E}_{X \sim q}[Z_t(X)]) \leq \frac{\text{KL}(q \| p_1)}{\eta} + 2\eta \sum_{t=1}^T \mathbb{E}_{X \sim p_t}[Z_t(X)^2] + \frac{T\varepsilon_p}{\eta}.$$

Since $\varepsilon_p = \eta/T$, the last term equals one. $\square$

Lemma 7. Under [Assumption 1](#), let $\rho \in (0, 1)$, and let $(q_t)_{t=1}^T$ be a predictable sequence of distributions in $\Delta(\mathcal{X})$. For every $t \in [T]$, define

$$X_t(q_t) := \mathbb{E}_{X \sim q_t}[\langle X, \ell_t \rangle], \quad \widehat{X}_t(q_t) := \mathbb{E}_{X \sim q_t}[\widehat{y}_t(X)].$$

Then, with probability at least $1 - \rho$,

$$\left| \sum_{t=1}^T (\widehat{X}_t(q_t) - X_t(q_t)) \right| \leq \frac{\eta}{4} \sum_{t=1}^T \mathbb{E}_{X \sim q_t}[c_t^\top X] + \frac{\log(2/\rho)}{\eta}.$$

Proof. For every $t \in [T]$, define

$$\Delta_t := \widehat{X}_t(q_t) - X_t(q_t).$$

Recall that $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_{t-1}]$. Since p_t , q_t , and ℓ_t are predictable, they are fixed conditionally on $\mathcal{F}_{t-1}$. The only randomness at round t comes from $X_t \sim p_t$. We first show that Δ_t has conditional mean zero. By the definition of the KW loss estimate,

$$\mathbb{E}_t[\widehat{\ell}_t] = M_t^{-1} \mathbb{E}_t[X_t X_t^\top] \ell_t = M_t^{-1} M_t \ell_t = \ell_t.$$

Therefore,

$$\mathbb{E}_t[\widehat{X}_t(q_t)] = X_t(q_t), \quad \mathbb{E}_t[\Delta_t] = 0.$$

Define $\bar{x}_t := \mathbb{E}_{X \sim q_t}[X]$. Then

$$\widehat{X}_t(q_t) = \bar{x}_t^\top M_t^{-1} X_t Y_t.$$

Since $|Y_t| \leq 1$,

$$\mathbb{E}_t[\widehat{X}_t(q_t)^2] \leq \bar{x}_t^\top M_t^{-1} \mathbb{E}_t[X_t X_t^\top] M_t^{-1} \bar{x}_t = \bar{x}_t^\top M_t^{-1} \bar{x}_t.$$

The matrix M_t^{-1} is positive semidefinite. Hence, Jensen's inequality gives

$$\bar{x}_t^\top M_t^{-1} \bar{x}_t \leq \mathbb{E}_{X \sim q_t} [X^\top M_t^{-1} X].$$

By Lemma 4 and $c_t^\top X = 4\phi_t(X)$,

$$X^\top M_t^{-1} X \leq \frac{c_t^\top X}{4}.$$

It follows that

$$\mathbb{E}_t[\widehat{X}_t(q_t)^2] \leq \frac{1}{4} \mathbb{E}_{X \sim q_t}[c_t^\top X].$$

Since $\mathbb{E}_t[\widehat{X}_t(q_t)] = X_t(q_t)$, we obtain

$$\mathbb{E}_t[\Delta_t^2] \leq \frac{1}{4} \mathbb{E}_{X \sim q_t}[c_t^\top X]. \quad (25)$$

We next bound Δ_t uniformly. Since $c_t^\top x \leq 4B$ for every $x \in \mathcal{X}$, the previous leverage bound gives

$$\bar{x}_t^\top M_t^{-1} \bar{x}_t \leq B, \quad X_t^\top M_t^{-1} X_t \leq B.$$

Using Cauchy–Schwarz in the norm induced by M_t^{-1} , we obtain

$$|\widehat{X}_t(q_t)| = \left| \bar{x}_t^\top M_t^{-1} X_t Y_t \right| \leq \sqrt{\bar{x}_t^\top M_t^{-1} \bar{x}_t} \sqrt{X_t^\top M_t^{-1} X_t} \leq B.$$

The action-normalized condition gives $|X_t(q_t)| \leq 1$. Since $B \geq 1$,

$$|\Delta_t| \leq B + 1 \leq 2B.$$

Moreover,

$$\eta B = \eta + \frac{8\eta d}{\lambda} = \eta + \frac{1}{16} \leq \frac{5}{64}.$$

Therefore,

$$|\eta \Delta_t| \leq 2\eta B \leq \frac{5}{32} < 1. \quad (26)$$

For every u with $|u| \leq 1$,

$$e^u \leq 1 + u + u^2.$$

Using (26), $\mathbb{E}_t[\Delta_t] = 0$, and (25), we obtain

$$\begin{aligned} \mathbb{E}_t[\exp(\eta \Delta_t)] &\leq 1 + \eta^2 \mathbb{E}_t[\Delta_t^2] \\ &\leq \exp(\eta^2 \mathbb{E}_t[\Delta_t^2]) \\ &\leq \exp\left(\frac{\eta^2}{4} \mathbb{E}_{X \sim q_t}[c_t^\top X]\right). \end{aligned}$$

For $s \in \{0, \dots, T\}$, define

$$\mathcal{M}_s := \exp\left(\eta \sum_{t=1}^s \Delta_t - \frac{\eta^2}{4} \sum_{t=1}^s \mathbb{E}_{X \sim q_t}[c_t^\top X]\right).$$

The previous conditional moment bound shows that $(\mathcal{M}_s)_{s=0}^T$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$. Hence, $\mathbb{E}[\mathcal{M}_T] \leq 1$. By Markov's inequality, with probability at least $1 - \rho/2$,

$$\sum_{t=1}^T \Delta_t \leq \frac{\eta}{4} \sum_{t=1}^T \mathbb{E}_{X \sim q_t}[c_t^\top X] + \frac{\log(2/\rho)}{\eta}.$$

The same argument applied to $-\Delta_t$ gives, with probability at least $1 - \rho/2$,

$$-\sum_{t=1}^T \Delta_t \leq \frac{\eta}{4} \sum_{t=1}^T \mathbb{E}_{X \sim q_t} [c_t^\top X] + \frac{\log(2/\rho)}{\eta}.$$

A union bound over the two events proves the result. $\square$

Lemma 8. Under [Assumption 1](#), for every $\rho \in (0, 1)$, with probability at least $1 - \rho$,

$$\sum_{t=1}^T \phi_t(X_t) \leq 2(2d+1)T + 2 \left(1 + \frac{8d}{\lambda}\right) \log \frac{1}{\rho}.$$

Proof. Recall that $B = 1 + 8d/\lambda$. Conditionally on $\mathcal{F}_{t-1}$, the distribution p_t and the function ϕ_t are fixed, while $X_t \sim p_t$. Therefore, [Lemma 4](#) gives

$$\mathbb{E}_t[\phi_t(X_t)] = \mathbb{E}_{X \sim p_t}[\phi_t(X)] = 2d+1, \quad 0 \leq \phi_t(X_t) \leq B.$$

Since $\phi_t(X_t)/B \in [0, 1]$, convexity of the exponential gives

$$e^u \leq 1 + (e-1)u, \quad u \in [0, 1].$$

It follows that

$$\mathbb{E}_t \left[\exp \left(\frac{\phi_t(X_t)}{B} \right) \right] \leq 1 + \frac{(e-1)(2d+1)}{B} \leq \exp \left(\frac{(e-1)(2d+1)}{B} \right).$$

For $s \in \{0, \dots, T\}$, define

$$\mathcal{M}_s := \exp \left(\frac{1}{B} \sum_{t=1}^s \phi_t(X_t) - \frac{(e-1)(2d+1)s}{B} \right).$$

The previous conditional moment bound shows that $(\mathcal{M}_s)_{s=0}^T$ is a nonnegative supermartingale with $\mathcal{M}_0 = 1$. Hence,

$$\mathbb{E}[\mathcal{M}_T] \leq 1.$$

By Markov's inequality, with probability at least $1 - \rho$,

$$\sum_{t=1}^T \phi_t(X_t) \leq (e-1)(2d+1)T + B \log \frac{1}{\rho}.$$

Using $e-1 \leq 2$ and substituting the definition of B proves the result. $\square$

Lemma 9. Under [Assumption 1](#), for every $\rho \in (0, 1)$, with probability at least $1 - \rho$,

$$\sum_{t=1}^T (\langle X_t, \ell_t \rangle - \mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle]) \leq \sqrt{2T \log \frac{1}{\rho}}.$$

Proof. Define

$$D_t := \langle X_t, \ell_t \rangle - \mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle].$$

Since p_t and ℓ_t are determined before X_t is sampled,

$$\mathbb{E}[D_t \mid \mathcal{F}_{t-1}] = 0.$$

Thus, $(D_t)_{t=1}^T$ is a martingale difference sequence.

By the action-normalized condition, $\langle X_t, \ell_t \rangle \in [-1, 1]$. Therefore, conditionally on $\mathcal{F}_{t-1}$, D_t belongs to an interval of length at most 2. The Azuma–Hoeffding inequality then gives, for every $u > 0$,

$$\mathbb{P}\left(\sum_{t=1}^T D_t \geq u\right) \leq \exp\left(-\frac{u^2}{2T}\right).$$

Taking $u = \sqrt{2T \log(1/\rho)}$ proves the result. $\square$

B.1 Proof of the main regret theorem

Theorem 1 (High-probability KW bound). *There exists a universal constant $C > 0$ such that the following holds. Let $T \geq 1$ be an integer, let $\delta \in (0, 1)$, and assume $1 \leq m \leq d - 1$. Run [Algorithm 1](#). Then, with probability at least $1 - \delta$,*

$$R_T \leq C \sqrt{dT \left(\log K + \log \frac{1}{\delta} \right)},$$

where $K = \binom{d}{m}$. One may take $C = 160$.

Proof. For every $x \in \mathcal{X}$, define the smoothed comparator

$$q_x := (1 - \lambda)\delta_x + \lambda U,$$

where δ_x assigns probability one to x , and U is the uniform distribution over $\mathcal{X}$. If $x_i = 1$, the i -th marginal of q_x is

$$1 - \lambda(1 - r),$$

while, if $x_i = 0$, it is λr . Therefore, $q_x \in \mathcal{P}_\lambda$. Moreover, since $p_1 = U$,

$$\text{KL}(q_x \| p_1) = \text{KL}(q_x \| U) = \log K - H(q_x) \leq \log K.$$

Applying [Lemma 6](#) with $q = q_x$ gives

$$\sum_{t=1}^T (\mathbb{E}_{X \sim p_t}[Z_t(X)] - \mathbb{E}_{X \sim q_x}[Z_t(X)]) \leq \frac{\log K}{\eta} + 1 + 2\eta \sum_{t=1}^T \mathbb{E}_{X \sim p_t}[Z_t(X)^2]. \quad (27)$$

We first bound the second-order term. Since $Z_t(X) = \hat{y}_t(X) - \eta c_t^\top X$, we have

$$Z_t(X)^2 \leq 2\hat{y}_t(X)^2 + 2\eta^2(c_t^\top X)^2.$$

Furthermore, we have

$$\begin{aligned} \mathbb{E}_{X \sim p_t}[\hat{y}_t(X)^2] &= Y_t^2 X_t^\top M_t^{-1} \mathbb{E}_{X \sim p_t}[X X^\top] M_t^{-1} X_t \\ &= Y_t^2 X_t^\top M_t^{-1} X_t \\ &\leq \phi_t(X_t), \end{aligned}$$

where the last inequality follows from $|Y_t| \leq 1$ and [Lemma 4](#). Recall that

$$0 \leq c_t^\top X \leq 4B, \quad \mathbb{E}_{X \sim p_t}[c_t^\top X] = 4(2d + 1).$$

Hence,

$$\mathbb{E}_{X \sim p_t}[(c_t^\top X)^2] \leq 4B\mathbb{E}_{X \sim p_t}[c_t^\top X] = 16B(2d+1).$$

It follows that

$$2\eta\mathbb{E}_{X \sim p_t}[Z_t(X)^2] \leq 4\eta\phi_t(X_t) + 64\eta^3B(2d+1). \quad (28)$$

Using $Z_t(X) = \hat{y}_t(X) - \eta c_t^\top X$ in (27), and then applying (28), gives

$$\begin{aligned} & \sum_{t=1}^T (\mathbb{E}_{X \sim p_t}[\hat{y}_t(X)] - \mathbb{E}_{X \sim q_x}[\hat{y}_t(X)]) \\ & \leq \frac{\log K}{\eta} + 1 + 4\eta \sum_{t=1}^T \phi_t(X_t) + 64\eta^3B(2d+1)T + 4\eta(2d+1)T - \eta \sum_{t=1}^T \mathbb{E}_{X \sim q_x}[c_t^\top X]. \end{aligned} \quad (29)$$

We now apply Lemma 8 with $\rho = \delta/4$. With probability at least $1 - \delta/4$,

$$4\eta \sum_{t=1}^T \phi_t(X_t) \leq 8\eta(2d+1)T + 8\eta B \log \frac{4}{\delta}.$$

Since

$$\eta B = \eta + \frac{8\eta d}{\lambda} = \eta + \frac{1}{16} \leq \frac{5}{64},$$

we have $8\eta B \leq 5/8$. Therefore,

$$4\eta \sum_{t=1}^T \phi_t(X_t) \leq 8\eta(2d+1)T + \log \frac{4}{\delta}. \quad (30)$$

Next, apply Lemma 7 to the predictable sequence $q_t = p_t$, with $\rho = \delta/4$. With probability at least $1 - \delta/4$,

$$\sum_{t=1}^T (\mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle] - \mathbb{E}_{X \sim p_t}[\hat{y}_t(X)]) \leq \eta(2d+1)T + \frac{\log(8/\delta)}{\eta}. \quad (31)$$

For every $x \in \mathcal{X}$, apply the same lemma to the constant predictable sequence $q_t = q_x$, with $\rho = \delta/(4K)$. A union bound over the K actions shows that, with probability at least $1 - \delta/4$, simultaneously for every $x \in \mathcal{X}$,

$$\sum_{t=1}^T (\mathbb{E}_{X \sim q_x}[\hat{y}_t(X)] - \mathbb{E}_{X \sim q_x}[\langle X, \ell_t \rangle]) \leq \frac{\eta}{4} \sum_{t=1}^T \mathbb{E}_{X \sim q_x}[c_t^\top X] + \frac{\log(8K/\delta)}{\eta}. \quad (32)$$

Combining (29), (30), (31), and (32), we obtain, simultaneously for every $x \in \mathcal{X}$,

$$\begin{aligned} & \sum_{t=1}^T (\mathbb{E}_{X \sim p_t}[\langle X, \ell_t \rangle] - \mathbb{E}_{X \sim q_x}[\langle X, \ell_t \rangle]) \leq \frac{\log K + \log(8/\delta) + \log(8K/\delta)}{\eta} \\ & + 1 + 13\eta(2d+1)T + 64\eta^3B(2d+1)T + \log \frac{4}{\delta} - \eta \sum_{t=1}^T \mathbb{E}_{X \sim q_x}[c_t^\top X] + \frac{\eta}{4} \sum_{t=1}^T \mathbb{E}_{X \sim q_x}[c_t^\top X]. \end{aligned} \quad (33)$$

The last two terms satisfy

$$-\eta \sum_{t=1}^T \mathbb{E}_{X \sim q_x} [c_t^\top X] + \frac{\eta}{4} \sum_{t=1}^T \mathbb{E}_{X \sim q_x} [c_t^\top X] = -\frac{3\eta}{4} \sum_{t=1}^T \mathbb{E}_{X \sim q_x} [c_t^\top X] \leq 0,$$

because $c_t^\top X \geq 0$. Moreover, $64\eta^3 B = 64\eta^2(\eta B) \leq 5\eta^2 \leq 5\eta$. Therefore, (33) gives

$$\sum_{t=1}^T (\mathbb{E}_{X \sim p_t} [\langle X, \ell_t \rangle] - \mathbb{E}_{X \sim q_x} [\langle X, \ell_t \rangle]) \leq \frac{\log K + \log(8/\delta) + \log(8K/\delta)}{\eta} + 1 + 18\eta(2d+1)T + \log \frac{4}{\delta}.$$

Using $2d+1 \leq 3d$ and enlarging the logarithmic terms, we obtain

$$\sum_{t=1}^T (\mathbb{E}_{X \sim p_t} [\langle X, \ell_t \rangle] - \mathbb{E}_{X \sim q_x} [\langle X, \ell_t \rangle]) \leq \frac{3 \log K + 3 \log(12/\delta)}{\eta} + 1 + 54\eta dT + \log \frac{4}{\delta}. \quad (34)$$

Let x^* be a best action in hindsight. Since $q_{x^*} = (1-\lambda)\delta_{x^*} + \lambda U$ and every action loss belongs to $[-1, 1]$,

$$\mathbb{E}_{X \sim q_{x^*}} [\langle X, \ell_t \rangle] \leq \langle x^*, \ell_t \rangle + 2\lambda.$$

Applying (34) with $x = x^*$ gives

$$\begin{aligned} & \sum_{t=1}^T (\mathbb{E}_{X \sim p_t} [\langle X, \ell_t \rangle] - \langle x^*, \ell_t \rangle) \\ & \leq \frac{3 \log K + 3 \log(12/\delta)}{\eta} + 1 + 54\eta dT + 2\lambda T + \log \frac{4}{\delta} \\ & \leq \frac{3 \log K + 3 \log(12/\delta)}{\eta} + 1 + 310\eta dT + \log \frac{4}{\delta}, \end{aligned}$$

where the last inequality uses $\lambda = 128\eta d$. Finally, apply Lemma 9 with $\rho = \delta/4$. With probability at least $1 - \delta/4$,

$$\sum_{t=1}^T (\langle X_t, \ell_t \rangle - \mathbb{E}_{X \sim p_t} [\langle X, \ell_t \rangle]) \leq \sqrt{2T \log \frac{4}{\delta}} \leq 2\sqrt{T \log \frac{4}{\delta}}.$$

Since $310 \leq 320$ and $1 + \log(4/\delta) \leq 4 \log(12/\delta)$, a union bound over the four failure events gives

$$R_T \leq \frac{3 \log K + 3 \log(12/\delta)}{\eta} + 320\eta dT + 4 \log \frac{12}{\delta} + 2\sqrt{T \log \frac{12}{\delta}}. \quad (35)$$

It remains to substitute the value of η . Define

$$A := \log K + \log \frac{12}{\delta}.$$

Also set

$$S := \log K + \log \frac{1}{\delta}.$$

Since $K = \binom{d}{m} \geq 2$, we have $S \geq \log 2$, and therefore

$$A = S + \log 12 \leq \left(1 + \frac{\log 12}{\log 2}\right) S \leq \frac{23}{5} S, \quad (36)$$

where the last inequality follows from $12^5 < 2^{18}$. If $\eta = \sqrt{A/(320dT)}$, then the first two terms in (35) are exactly

$$\frac{3A}{\eta} + 320\eta dT = 4\sqrt{320dT A}.$$

Moreover,

$$4 \log \frac{12}{\delta} \leq 4A, \quad 2\sqrt{T \log \frac{12}{\delta}} \leq 2\sqrt{dT A}.$$

It remains to consider the case in which $\eta = 1/(256d)$. In this case,

$$\sqrt{\frac{A}{320dT}} \geq \frac{1}{256d},$$

which implies

$$T \leq \frac{256^2}{320} dA.$$

Since the action-normalized condition gives the deterministic bound $R_T \leq 2T$, we also have

$$R_T \leq 2\sqrt{\frac{256^2}{320}} \sqrt{dT A} \leq 29\sqrt{dT A}.$$

In the non-clipped case, Equation (35) is at most $74\sqrt{dT A} + 4A$. Since $29 < 74$ and $A > 0$, the same bound also holds in the clipped case. Hence, by (36),

$$R_T \leq 74\sqrt{\frac{23}{5}} \sqrt{dT S} + \frac{92}{5} S. \quad (37)$$

Set

$$\gamma := \sqrt{\frac{S}{dT}} > 0.$$

Since

$$S = \gamma \sqrt{dT S},$$

if $\gamma \leq 1/160$, then

$$\begin{aligned} R_T &\leq \left(74\sqrt{\frac{23}{5}} + \frac{92}{5}\gamma \right) \sqrt{dT S} \\ &\leq \left(74\frac{43}{20} + \frac{92}{5}\frac{1}{160} \right) \sqrt{dT S} \\ &= \frac{31843}{200} \sqrt{dT S} < 160\sqrt{dT S}, \end{aligned}$$

where we used

$$\sqrt{\frac{23}{5}} < \frac{43}{20}.$$

If instead $\gamma > 1/160$, the deterministic bound $R_T \leq 2T$ and $d \geq 2$ give

$$R_T \leq 2T = \frac{2}{d\gamma} \sqrt{dT S} \leq \frac{1}{\gamma} \sqrt{dT S} < 160\sqrt{dT S}.$$

Thus, in all cases,

$$R_T \leq 160\sqrt{dT \left(\log K + \log \frac{1}{\delta} \right)},$$

which proves the theorem with $C = 160$. □

C Polynomial-time implementation

This appendix proves [Proposition 1](#). We show that our algorithm can be implemented in polynomial time without enumerating $\mathcal{X}$. It maintains each p_t as a weighted m -set distribution, computes the required moments and draws samples in polynomial time, and reduces the target KL projection to a convex problem in $2d$ variables. The convex problem is solved only to the inverse-polynomial accuracy specified below.

C.1 Updates and KL projection

We first identify the exact KL projection toward which the algorithm computes and show that it reduces to a convex problem in $2d$ variables. We then prove that solving this problem to an explicit objective accuracy is sufficient for the approximate update in [Algorithm 1](#).

Suppose that $p_t = p_{\theta_t}$. Define the unprojected multiplicative-weights update

$$\tilde{p}_{t+1}(x) := \frac{p_t(x) \exp(-\eta Z_t(x))}{\mathbb{E}_{X \sim p_t}[\exp(-\eta Z_t(X))]}.$$

For every $p \in \mathcal{P}_\lambda$,

$$\eta \mathbb{E}_{X \sim p}[Z_t(X)] + \text{KL}(p \| p_t) = \text{KL}(p \| \tilde{p}_{t+1}) - \log \mathbb{E}_{X \sim p_t}[\exp(-\eta Z_t(X))].$$

The last term does not depend on p . Therefore, the exact constrained OMD iterate is the KL projection of $\tilde{p}_{t+1}$ onto $\mathcal{P}_\lambda$, namely,

$$p_{t+1}^* \in \arg \min_{p \in \mathcal{P}_\lambda} \text{KL}(p \| \tilde{p}_{t+1}).$$

The algorithm computes the approximate replacement specified in [Algorithm 1](#) rather than assuming access to this exact projection.

We next show that the first step preserves the weighted m -set form. Since $c_t^\top x$ is affine in x , the surrogate loss is affine on $\mathcal{X}$:

$$Z_t(x) = \langle x, \hat{\ell}_t \rangle - \eta c_t^\top x = z_t^\top x, \quad z_t := \hat{\ell}_t - \eta c_t.$$

Consequently,

$$\tilde{p}_{t+1}(x) = \frac{p_{\theta_t}(x) \exp(-\eta z_t^\top x)}{\mathbb{E}_{X \sim p_{\theta_t}}[\exp(-\eta z_t^\top X)]} = p_{\tilde{\theta}_{t+1}}(x), \quad \tilde{\theta}_{t+1} := \theta_t - \eta z_t. \quad (38)$$

Hence, if p_t is a weighted m -set distribution, then so is $\tilde{p}_{t+1}$.

It remains to analyze the target KL projection. The next lemma shows that the projection of a weighted m -set distribution onto $\mathcal{P}_\lambda$ is again a weighted m -set distribution. It also shows that the projection can be obtained by solving a convex problem in $2d$ variables.

Lemma 10. *Define*

$$a_i := \lambda r, \quad b_i := 1 - \lambda(1 - r), \quad i \in [d].$$

For every $\tilde{\theta} \in \mathbb{R}^d$, the KL projection

$$\arg \min_{p \in \mathcal{P}_\lambda} \text{KL}(p \| p_{\tilde{\theta}})$$

is a weighted m -set distribution p_{θ^+} , where $\theta^+ = \tilde{\theta} + \alpha^* - \beta^*$. Here, $(\alpha^*, \beta^*) \in \mathbb{R}_+^d \times \mathbb{R}_+^d$ minimizes the convex function

$$\Psi_{\tilde{\theta}}(\alpha, \beta) := F(\tilde{\theta} + \alpha - \beta) - \alpha^\top a + \beta^\top b.$$

Moreover,

$$\frac{\partial \Psi_{\tilde{\theta}}}{\partial \alpha_i} = \mu_i(\tilde{\theta} + \alpha - \beta) - a_i, \quad \frac{\partial \Psi_{\tilde{\theta}}}{\partial \beta_i} = b_i - \mu_i(\tilde{\theta} + \alpha - \beta).$$

Proof. The projection minimizes $\text{KL}(p \| p_{\tilde{\theta}})$ over $p \in \Delta(\mathcal{X})$, subject to

$$a_i - \mathbb{E}_{X \sim p}[X_i] \leq 0, \quad \mathbb{E}_{X \sim p}[X_i] - b_i \leq 0.$$

The uniform distribution has all marginals equal to r , strictly between a_i and b_i . Hence, Slater's condition holds. We may therefore use the standard strong-duality and KKT results of Boyd and Vandenberghe [7, Sections 5.2.3 and 5.5.3].

Introduce multipliers $\alpha_i, \beta_i \geq 0$. The Lagrangian is

$$\mathcal{L}(p, \alpha, \beta) = \text{KL}(p \| p_{\tilde{\theta}}) + \alpha^\top a - \beta^\top b + \mathbb{E}_{X \sim p}[(\beta - \alpha)^\top X].$$

For fixed (α, β) , minimizing over p gives

$$p(x) \propto p_{\tilde{\theta}}(x) e^{(\alpha - \beta)^\top x} = p_{\tilde{\theta} + \alpha - \beta}(x).$$

The resulting dual maximization is equivalent to minimizing $\Psi_{\tilde{\theta}}$. Strong duality proves the first claim, and the derivative identities follow from $\nabla F = \mu$. $\square$

Lemma 11. Let $0 < \lambda \leq 1/2$, let $\varepsilon > 0$, and let $\tilde{\theta} \in \mathbb{R}^d$ satisfy

$$\text{KL}(U \| p_{\tilde{\theta}}) \leq T.$$

Define

$$h := \frac{\lambda}{2} \min\{r, 1 - r\} = \frac{\lambda s}{2d}, \quad R := 2dT,$$

and

$$\tau := \min \left\{ 2h^2, \frac{\varepsilon}{2}, \frac{\varepsilon^2}{2R^2} \right\}. \quad (39)$$

Let

$$\mathcal{D}_R := \left\{ (\alpha, \beta) \in \mathbb{R}_+^d \times \mathbb{R}_+^d : \|\alpha\|_1 + \|\beta\|_1 \leq R \right\}.$$

Suppose that $(\alpha, \beta) \in \mathcal{D}_R$ satisfies

$$\Psi_{\tilde{\theta}}(\alpha, \beta) \leq \min_{(u, v) \in \mathcal{D}_R} \Psi_{\tilde{\theta}}(u, v) + \tau. \quad (40)$$

Set

$$\theta^+ := \tilde{\theta} + \alpha - \beta, \quad p^+ := p_{\theta^+}.$$

Then $p^+ \in \mathcal{P}_{\lambda/2}$, and, for every $q \in \mathcal{P}_\lambda$,

$$\text{KL}(q \| p^+) + \text{KL}(p^+ \| p_{\tilde{\theta}}) \leq \text{KL}(q \| p_{\tilde{\theta}}) + \varepsilon. \quad (41)$$

Moreover, a pair satisfying (40) can be computed by the ellipsoid method in time

$$\text{poly} \left(d, m, \log R, \log \frac{1}{\tau} \right).$$

Proof. Write

$$a_i := \lambda r, \quad b_i := 1 - \lambda(1 - r).$$

Let (α^*, β^*) be an exact minimizer of $\Psi_{\tilde{\theta}}$, let

$$\theta^* := \tilde{\theta} + \alpha^* - \beta^*, \quad p^* := p_{\theta^*},$$

and write $\mu^* := \mu(p^*)$. We first show that an exact minimizer belongs to $\mathcal{D}_R$. By strong duality, the value of the dual function at (α^*, β^*) equals the value of the KL projection problem and is therefore nonnegative. Evaluating the Lagrangian from the proof of [Lemma 10](#) at the uniform distribution gives

$$0 \leq \text{KL}(U \| p_{\tilde{\theta}}) - (1 - \lambda)r \|\alpha^*\|_1 - (1 - \lambda)(1 - r) \|\beta^*\|_1.$$

Consequently,

$$\|\alpha^*\|_1 + \|\beta^*\|_1 \leq \frac{T}{(1 - \lambda) \min\{r, 1 - r\}} \leq \frac{2dT}{s} \leq R.$$

Thus the minimum over $\mathcal{D}_R$ in (40) is the global minimum.

Let $p := p^+$, let $\mu := \mu(p)$, and abbreviate $\Psi := \Psi_{\tilde{\theta}}$. The KKT conditions for the exact projection give

$$a_i \leq \mu_i^* \leq b_i, \quad \alpha_i^*(\mu_i^* - a_i) = 0, \quad \beta_i^*(b_i - \mu_i^*) = 0.$$

Using

$$\text{KL}(p^* \| p) = F(\theta^+) - F(\theta^*) - (\mu^*)^\top (\theta^+ - \theta^*),$$

and the complementary-slackness identities above, direct expansion gives the exact identity

$$\Psi(\alpha, \beta) - \Psi(\alpha^*, \beta^*) = \text{KL}(p^* \| p) + \alpha^\top (\mu^* - a) + \beta^\top (b - \mu^*). \quad (42)$$

All three terms on the right-hand side are nonnegative. Hence,

$$\text{KL}(p^* \| p) \leq \tau.$$

Pinsker's inequality and the fact that every coordinate of an action belongs to $[0, 1]$ imply

$$\|\mu - \mu^*\|_\infty \leq \text{TV}(p, p^*) \leq \sqrt{\frac{\tau}{2}} \leq h.$$

Every marginal of p^* belongs to $[\lambda r, 1 - \lambda(1 - r)]$. The distance from this interval to the corresponding boundary of $[\lambda r/2, 1 - \lambda(1 - r)/2]$ is at least h . Therefore, $p \in \mathcal{P}_{\lambda/2}$.

Define the residual

$$\kappa := \alpha^\top (\mu - a) + \beta^\top (b - \mu).$$

By (42) and $(\alpha, \beta) \in \mathcal{D}_R$,

$$\begin{aligned} \kappa &= \alpha^\top (\mu^* - a) + \beta^\top (b - \mu^*) + (\alpha - \beta)^\top (\mu - \mu^*) \\ &\leq \tau + R \sqrt{\frac{\tau}{2}} \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon, \end{aligned}$$

where the last inequality follows from (39).

For every $q \in \mathcal{P}_\lambda$, the exponential-family representation gives the three-point identity

$$\begin{aligned} \text{KL}(q \| p_{\tilde{\theta}}) - \text{KL}(q \| p) - \text{KL}(p \| p_{\tilde{\theta}}) \\ = (\mu(q) - \mu)^\top (\alpha - \beta). \end{aligned}$$

Since $a \leq \mu(q) \leq b$ coordinatewise and $\alpha, \beta \geq 0$, the right-hand side is at least $-\kappa$. The bound $\kappa \leq \varepsilon$ proves (41).

It remains to justify the computational claim. Put

$$L := \sqrt{2d}, \quad r_0 := \frac{R}{8d}, \quad w_0 := \frac{R}{4d} \mathbf{1}_{2d}.$$

For $v \in \mathbb{R}^{2d}$ and $r > 0$, write $B(v, r) := \{u \in \mathbb{R}^{2d} : \|u - v\|_2 \leq r\}$. The set $\mathcal{D}_R$ has an explicit separation procedure, is contained in the Euclidean ball of radius R , and contains $B(w_0, r_0)$. Indeed, every point of this ball has nonnegative coordinates and ℓ_1 -norm at most $R/2 + \sqrt{2d}r_0 \leq R$. By the derivative formulas in Lemma 10,

$$\|\nabla \Psi_{\tilde{\theta}}(w)\|_2 \leq L \quad \text{for every } w \in \mathbb{R}^{2d}.$$

Thus $\Psi_{\tilde{\theta}}$ is globally L -Lipschitz. Its value and gradient can be evaluated using $\text{poly}(d, m)$ operations by Lemma 12.

Let w^* minimize $\Psi_{\tilde{\theta}}$ over $\mathcal{D}_R$. For $0 < \zeta \leq r_0$, define

$$w_\zeta := \left(1 - \frac{\zeta}{r_0}\right) w^* + \frac{\zeta}{r_0} w_0.$$

Convexity of $\mathcal{D}_R$ and $B(w_0, r_0) \subseteq \mathcal{D}_R$ imply $B(w_\zeta, \zeta) \subseteq \mathcal{D}_R$. Moreover, $\|w^* - w_0\|_2 \leq 2R$, and hence

$$\Psi_{\tilde{\theta}}(w_\zeta) \leq \Psi_{\tilde{\theta}}(w^*) + 16dL\zeta.$$

Set

$$\zeta := \min \left\{ r_0, \frac{\tau}{1 + (16d + 1)L} \right\}.$$

Weak constrained convex-function minimization returns a point $\hat{w}$ at Euclidean distance at most ζ from $\mathcal{D}_R$, whose objective is at most ζ above the objective of every point whose closed ζ -ball is contained in $\mathcal{D}_R$ [17, Problem (2.1.22) and Theorem 4.3.13]. In particular,

$$\Psi_{\tilde{\theta}}(\hat{w}) \leq \Psi_{\tilde{\theta}}(w^*) + (16dL + 1)\zeta.$$

Let $\bar{w}$ be the Euclidean projection of $\hat{w}$ onto $\mathcal{D}_R$. Since $\mathcal{D}_R$ is the nonnegative ℓ_1 -ball, its coordinates have the form

$$\bar{w}_i = \max\{\hat{w}_i - \chi, 0\},$$

where $\chi = 0$ if the positive part of $\hat{w}$ has ℓ_1 -norm at most R , and otherwise χ is chosen so that $\sum_i \max\{\hat{w}_i - \chi, 0\} = R$. Sorting the coordinates finds χ in $O(d \log d)$ time. Furthermore, $\|\bar{w} - \hat{w}\|_2 \leq \zeta$, so global Lipschitzness gives

$$\begin{aligned} \Psi_{\tilde{\theta}}(\bar{w}) &\leq \Psi_{\tilde{\theta}}(w^*) + (1 + (16d + 1)L)\zeta \\ &\leq \Psi_{\tilde{\theta}}(w^*) + \tau. \end{aligned}$$

Writing $\bar{w} = (\alpha, \beta)$ gives a pair in $\mathcal{D}_R$ satisfying (40). Since $R/r_0 = 8d$ and $\log(1/\zeta) = O(\log d + \log R + \log(1/\tau))$, the weak optimizer and the final projection use $\text{poly}(d, m, \log R, \log(1/\tau))$ operations. $\square$

C.2 Moment and sampling routines

We now show how to compute the moments of a weighted m -set distribution and draw samples from it without enumerating $\mathcal{X}$. Given its parameters θ , we compute its marginals, its second-moment matrix, and an exact sample using dynamic programs for elementary symmetric polynomials.

Lemma 12. *Given $\theta \in \mathbb{R}^d$, the log-partition function $F(\theta)$, all marginals $\mu_i(p_\theta)$, all pairwise marginals*

$$\pi_{ij}(p_\theta) := \mathbb{P}_{X \sim p_\theta}(X_i = X_j = 1),$$

and a sample from p_θ can be drawn using a number of arithmetic operations polynomial in d and m , without enumerating $\mathcal{X}$.

Proof. Let $w_i := e^{\theta_i}$, and define

$$E[i, k] := e_k(w_1, \dots, w_i).$$

The recurrence

$$E[i, k] = E[i-1, k] + w_i E[i-1, k-1] \quad (43)$$

computes all entries with $0 \leq i \leq d$ and $0 \leq k \leq m$ in $O(dm)$ arithmetic operations. In particular, $E[0, 0] = 1$ and $E[i, k] = 0$ whenever $k < 0$ or $k > i$. Moreover,

$$F(\theta) = \log E[d, m].$$

Write $E_k := e_k(w)$. For each $i \in [d]$, define

$$E_0^{(-i)} := 1, \quad E_k^{(-i)} := E_k - w_i E_{k-1}^{(-i)}.$$

For $i \neq j$, define

$$E_0^{(-i, -j)} := 1, \quad E_k^{(-i, -j)} := E_k^{(-i)} - w_j E_{k-1}^{(-i, -j)}.$$

We use the value zero for both arrays at negative indices. These recurrences give

$$\mu_i(p_\theta) = \frac{w_i E_{m-1}^{(-i)}}{E_m}, \quad \pi_{ii}(p_\theta) = \mu_i(p_\theta), \quad \pi_{ij}(p_\theta) = \frac{w_i w_j E_{m-2}^{(-i, -j)}}{E_m} \quad (i \neq j). \quad (44)$$

All pairwise marginals can be obtained in $O(d^2 m)$ arithmetic operations. The recurrence is Method 2 of Chen and Liu [12, Section 2].

For sampling, start from $(i, k) = (d, m)$. At state (i, k) , include item i with probability

$$\frac{w_i E[i-1, k-1]}{E[i, k]}. \quad (45)$$

If item i is selected, continue from $(i-1, k-1)$; otherwise, continue from $(i-1, k)$. The recurrence (43) shows that (45) is the correct conditional probability. Thus, the procedure samples from p_θ using $O(dm)$ arithmetic operations after the table has been computed. It is the reverse-indexed conditional-Bernoulli procedure of Chen and Liu [12, Section 4, Procedure 3]; it is also equivalent to their direct backward-path construction (Procedure 5) and to the diagonal- L specialization of Kulesza and Taskar [26, Section 3.1 and Algorithm 2]. $\square$

C.3 Proof of Proposition 1

Proposition 1. *Algorithm 1 can be implemented in $T \cdot \text{poly}(d, m, \log T)$ time and $\text{poly}(d, m)$ space, without enumerating $\mathcal{X}$. At every round, p_t is a weighted m -set distribution p_{θ_t} represented by the d parameters $\theta_{t,1}, \dots, \theta_{t,d}$.*

Proof. Recall that

$$\varepsilon_p = \frac{\eta}{T}.$$

We prove by induction that every p_t is a weighted m -set distribution in $\mathcal{P}_{\lambda/2}$ and that

$$\text{KL}(U \| p_t) \leq (t-1) \left(\frac{1}{2} + \varepsilon_p \right). \quad (46)$$

This holds at $t = 1$, because $\theta_1 = 0$ and $p_1 = p_{\theta_1} = U$.

Suppose that the claim holds at round t . Given θ_t , Lemma 12 computes the marginals and the second-moment matrix M_t in $O(d^2 m)$ operations. The same marginals determine ϕ_t and c_t in $O(d)$ operations. The sampling procedure in Lemma 12 draws X_t from p_t without enumerating $\mathcal{X}$: at each step, it draws one uniform random variable and compares it with the conditional probability in (45).

By Lemma 3, M_t is positive definite. Therefore, M_t^{-1} and $\hat{\ell}_t$ can be computed in $O(d^3)$ operations. The surrogate loss is represented by the d -dimensional vector $z_t = \hat{\ell}_t - \eta c_t$, and (38) gives $\tilde{p}_{t+1} = p_{\tilde{\theta}_{t+1}}$.

We next verify the hypothesis of Lemma 11. By the definition of the unprojected update,

$$\text{KL}(U \| \tilde{p}_{t+1}) = \text{KL}(U \| p_t) + \eta \mathbb{E}_{X \sim U} [Z_t(X)] + \log \mathbb{E}_{X \sim p_t} [e^{-\eta Z_t(X)}].$$

Under the induction hypothesis, Lemma 5 gives $|\eta Z_t(x)| \leq 1/4$ for every $x \in \mathcal{X}$. In particular,

$$\eta \mathbb{E}_{X \sim U} [Z_t(X)] \leq \frac{1}{4}, \quad \log \mathbb{E}_{X \sim p_t} [e^{-\eta Z_t(X)}] \leq \log(e^{1/4}) = \frac{1}{4}.$$

Hence,

$$\text{KL}(U \| \tilde{p}_{t+1}) \leq \text{KL}(U \| p_t) + \frac{1}{2} \leq \frac{t}{2} + (t-1)\varepsilon_p \leq T,$$

where the last inequality uses $t \leq T$ and $T\varepsilon_p = \eta \leq 1/64$.

Apply Lemma 11 with $\varepsilon = \varepsilon_p$. It computes $(\alpha_t, \beta_t) \in \mathcal{D}_{2dT}$ so that (40) holds with the value τ in (39). Set

$$\theta_{t+1} := \tilde{\theta}_{t+1} + \alpha_t - \beta_t, \quad p_{t+1} := p_{\theta_{t+1}}.$$

The lemma gives $p_{t+1} \in \mathcal{P}_{\lambda/2}$ and the KL inequality required in Algorithm 1. Since $U \in \mathcal{P}_\lambda$, that inequality also gives

$$\text{KL}(U \| p_{t+1}) \leq \text{KL}(U \| \tilde{p}_{t+1}) + \varepsilon_p \leq t \left(\frac{1}{2} + \varepsilon_p \right).$$

This closes the induction.

It remains to bound the number of operations uniformly over the rounds. Since $K \geq 2$ and $\delta < 1$, the learning rate in Algorithm 1 satisfies

$$\eta \geq \frac{1}{256d\sqrt{T}}.$$

Consequently,

$$\varepsilon_p \geq \frac{1}{256dT^{3/2}}, \quad \frac{\lambda s}{2d} \geq \frac{1}{4d\sqrt{T}}, \quad R = 2dT.$$

The value τ in (39) therefore satisfies

$$\tau \geq \frac{1}{2^{19}d^4T^5}.$$

Thus $\log R + \log(1/\tau) = O(\log(dT))$, and Lemma 11 runs in $\text{poly}(d, m, \log T)$ time per round. All other steps run in time polynomial in d and m . The algorithm therefore runs in $T \cdot \text{poly}(d, m, \log T)$ time, uses $\text{poly}(d, m)$ space, and never enumerates $\mathcal{X}$. $\square$

D Proof of the matching lower bound

We prove the following fully explicit version of Theorem 2.

Theorem 3. *Let $d \geq 2$, let $1 \leq m \leq d - 1$, and set*

$$s := \min\{m, d - m\}.$$

Let $\delta \in (0, 2^{-36}]$, and assume that

$$T \geq \left\lceil d^2 + 64ds \log \frac{ed}{s} + 4ds^2 \log \frac{1}{\delta} \right\rceil. \quad (47)$$

Then, for every randomized policy on $\mathcal{X}_{d,m}$, there exists a deterministic adaptive non-anticipating loss process whose action losses belong to $[0, 1]$ and such that

$$\mathbb{P} \left(R_T \geq 2^{-39} \sqrt{dT \left(\log \binom{d}{m} + \log \frac{1}{\delta} \right)} \right) \geq \delta.$$

Thus, Theorem 3 implies Theorem 2 by taking

$$c = 2^{-39}, \quad \delta_0 = 2^{-36}, \quad \mathfrak{T}(d, m, \delta) = \left\lceil d^2 + 64ds \log \frac{ed}{s} + 4ds^2 \log \frac{1}{\delta} \right\rceil.$$

Here $s := \min\{m, d - m\}$. Since $s \leq d$ and $\log(ed/s) \leq \log(ed) \leq d$ for $d \geq 2$, this threshold is at most

$$1 + d^2 + 64d^3 + 4d^3 \log \frac{1}{\delta},$$

and is therefore bounded by a polynomial in the required parameters. The rest of this appendix proves Theorem 3.

We first consider exact k -set actions in dimension n , where $1 \leq k \leq n/2$. We write

$$\mathcal{X}_{n,k} := \{x \in \{0, 1\}^n : \|x\|_1 = k\}, \quad L_\delta := \log \frac{1}{\delta}.$$

The action-count and confidence contributions are proved separately. For $t \in [T]$, let

$$\mathcal{H}_t := (X_1, Y_1, \dots, X_t, Y_t), \quad \mathcal{H}_0 := \emptyset,$$

denote the action–feedback history up to round t . For any probability law P , we write $P^{\mathcal{H}_t}$ for the induced law of $\mathcal{H}_t$. The action-count proof uses two constructions, according to whether $\lfloor n/k \rfloor$ is at least

$$r_\star := 2^{32}. \quad (48)$$

For probability measures Q and P , we use

$$\text{TV}(Q, P) := \sup_E |Q(E) - P(E)|.$$

When Q is absolutely continuous with respect to P , we also write

$$\text{KL}(Q\|P) := \int \log \left(\frac{dQ}{dP} \right) dQ, \quad \chi^2(Q\|P) := \int \left(\frac{dQ}{dP} - 1 \right)^2 dP;$$

otherwise these two quantities are $+\infty$. The same notation for densities means the corresponding probability laws.

Every hard instance below is first described using an auxiliary random table sampled independently of the learner. Once the table is fixed, the losses are deterministic functions of the past actions and feedback, and hence form a deterministic non-anticipating adversary. The following elementary observation will be used without further comment.

Lemma 13. *Let Ξ be an auxiliary random table, let R collect all the learner's private randomization, and suppose that Ξ and R are independent. Let E be a measurable event in the joint experiment. If $\mathbb{P}_{\Xi, R}(E) \geq p$, then there exists a deterministic table ξ such that*

$$\mathbb{P}_R(E \text{ when } \Xi = \xi) \geq p.$$

Proof. For each table ξ , let

$$q(\xi) := \mathbb{P}_R(E \text{ when } \Xi = \xi).$$

By the product structure and Tonelli's theorem,

$$\mathbb{P}_{\Xi, R}(E) = \int q(\xi) \mathbb{P}_{\Xi}(d\xi).$$

Since the integral is at least p , some ξ satisfies $q(\xi) \geq p$. This formulation uses sections of the event and remains valid when the law of Ξ is continuous. $\square$

The following symmetry lets us work only with sets of size at most half the dimension.

Lemma 14 (Complement symmetry). *Let $1 \leq s \leq d/2$ and $m := d - s$. Given a randomized policy on $\mathcal{X}_{d, m}$, couple it with a policy on $\mathcal{X}_{d, s}$, using the same private randomization, by setting*

$$y_t := \mathbf{1} - x_t$$

whenever the original policy selects x_t . If g_t is a deterministic non-anticipating loss process for the coupled s -set policy, define recursively

$$\tilde{\ell}_t := -g_t + \frac{\mathbf{1}^\top g_t}{m} \mathbf{1}. \tag{49}$$

Then the two policies receive the same scalar feedback at every round and, pathwise,

$$R_T^{(m)}(\tilde{\ell}_{1:T}) = R_T^{(s)}(g_{1:T}).$$

Moreover, $\tilde{\ell}_t$ is deterministic and non-anticipating, and all its m -set action losses belong to $[0, 1]$ whenever all s -set action losses of g_t belong to $[0, 1]$.

Proof. For every complementary pair $x = \mathbf{1} - y$ with $\|x\|_1 = m$ and $\|y\|_1 = s$, (49) gives

$$\langle x, \tilde{\ell}_t \rangle = -\langle x, g_t \rangle + \mathbf{1}^\top g_t = \langle y, g_t \rangle.$$

Starting from the empty history, induction on t therefore shows that the two policies receive the same feedback and continue to select complementary actions. The displayed identity also pairs every fixed comparator in $\mathcal{X}_{d,m}$ with one in $\mathcal{X}_{d,s}$, so the two regrets are equal pathwise. It also proves the assertion about the interval $[0, 1]$. Finally, the history through round $t - 1$ is the same in the two experiments; hence determinism and non-anticipation of g_t imply the corresponding properties of $\tilde{\ell}_t$. $\square$

D.1 Preliminary lemmas

Let

$$g(z) := 1 - z^2, \quad F_u(z) := \frac{z}{2} + ug(z), \quad z \in [-1, 1].$$

We now establish the density estimates used throughout the proof with numerical constants.

Lemma 15. *Set*

$$u_0 := \frac{1}{16}.$$

For every $|u| \leq u_0$, the map F_u is an increasing diffeomorphism from $[-1, 1]$ onto $[-1/2, 1/2]$. Let p_u be the density of $F_u(Z)$ when $Z \sim \text{Unif}[-1, 1]$. For every $|u|, |v| \leq u_0$,

$$\text{KL}(p_u \| p_v) \leq 128(u - v)^2, \quad (50)$$

$$\int_{-1/2}^{1/2} \frac{p_u(y)^2}{p_v(y)} dy \leq \exp(128(u - v)^2). \quad (51)$$

Moreover, writing

$$H_y(u) := \log F'_u(F_u^{-1}(y)), \quad G_y(u) := g(F_u^{-1}(y)),$$

we have

$$|H_y(u + s) + H_y(u + t) - H_y(u) - H_y(u + s + t)| \leq 128|st|, \quad (52)$$

$$|G'_y(u)| \leq 6G_y(u), \quad (53)$$

whenever all four parameters appearing in (52), or the parameter in (53), belong to $[-u_0, u_0]$.

Proof. We have

$$F'_u(z) = \frac{1}{2} - 2uz.$$

For $|u| \leq 1/16$ and $z \in [-1, 1]$,

$$\frac{3}{8} \leq F'_u(z) \leq \frac{5}{8}. \quad (54)$$

Since $F_u(-1) = -1/2$ and $F_u(1) = 1/2$, this proves the diffeomorphism claim.

Fix $y \in [-1/2, 1/2]$ and write $z = z(u, y) := F_u^{-1}(y)$. Implicit differentiation gives

$$\partial_u z = -\frac{g(z)}{F'_u(z)}, \quad |\partial_u z| \leq \frac{8}{3}.$$

The density is

$$p_u(y) = \frac{1}{2F'_u(z(u, y))}.$$

If $A(u) := F'_u(z(u, y))$, then

$$|A'(u)| = |-2z - 2u\partial_u z| \leq 2 + \frac{1}{3} = \frac{7}{3}.$$

Using (54),

$$|\partial_u p_u(y)| = \frac{|A'(u)|}{2A(u)^2} \leq \frac{224}{27} < 9.$$

Also $p_v(y) \geq 4/5$. Therefore, by the mean-value theorem,

$$\int_{-1/2}^{1/2} \frac{(p_u(y) - p_v(y))^2}{p_v(y)} dy \leq \frac{5}{4} 9^2 (u - v)^2 < 128(u - v)^2.$$

The inequality $\text{KL} \leq \chi^2$ proves (50). Furthermore,

$$\int \frac{p_u^2}{p_v} = 1 + \int \frac{(p_u - p_v)^2}{p_v} \leq 1 + 128(u - v)^2 \leq e^{128(u-v)^2},$$

which is (51).

For the remaining estimates, differentiating the inverse relation twice gives

$$\partial_{uu} z = \frac{4z\partial_u z + 2u(\partial_u z)^2}{F'_u(z)}, \quad |\partial_{uu} z| \leq \frac{832}{27} < 31.$$

Consequently,

$$|A''(u)| = |-4\partial_u z - 2u\partial_{uu} z| < 15.$$

Since $H_y(u) = \log A(u)$, we obtain

$$|H''_y(u)| \leq \frac{15}{3/8} + \left(\frac{7/3}{3/8}\right)^2 < 79 < 128.$$

More explicitly, the two-dimensional fundamental theorem of calculus gives

$$H_y(u + s) + H_y(u + t) - H_y(u) - H_y(u + s + t) = -st \int_0^1 \int_0^1 H''_y(u + \alpha s + \beta t) d\alpha d\beta.$$

Every argument in this integral is a convex combination of the four parameters in (52); hence it belongs to $[-u_0, u_0]$. The preceding bound proves (52). Finally,

$$G'_y(u) = -2z\partial_u z = \frac{2zg(z)}{F'_u(z)},$$

and hence, again by (54),

$$|G'_y(u)| \leq \frac{16}{3} G_y(u) \leq 6G_y(u).$$

□

We also need the following elementary probability facts.

Lemma 16. Let $E_1, \dots, E_N$ be events and $W_1, \dots, W_N$ be nonnegative random variables under a probability measure P . If, for some $p \in (0, 1]$ and $w \geq 0$,

$$\frac{1}{N} \sum_{i=1}^N P(E_i) \geq p, \quad \frac{1}{N} \sum_{i=1}^N \mathbb{E}_P[W_i] \leq w,$$

then some $i \in [N]$ satisfies

$$P(E_i) \geq \frac{p}{2}, \quad \mathbb{E}_P[W_i] \leq \frac{2w}{p}.$$

Proof. Let $G := \{i : P(E_i) \geq p/2\}$. Since probabilities are at most one,

$$p \leq \frac{|G|}{N} + \left(1 - \frac{|G|}{N}\right) \frac{p}{2},$$

so $|G| \geq pN/2$. Hence some $i \in G$ satisfies

$$\mathbb{E}_P[W_i] \leq \frac{Nw}{|G|} \leq \frac{2w}{p}. \quad \square$$

Lemma 17. Let E be an event under probability measures P, Q , and let $p \in (0, 1]$. If

$$P(E) \geq p, \quad \text{KL}(P\|Q) \leq \frac{p}{4} \log \frac{1}{\delta}, \quad 0 < \delta \leq 2^{-2/p}, \quad (55)$$

then $Q(E) \geq \delta$.

Proof. Suppose for contradiction that $Q(E) < \delta$. If $Q(E) = 0$, then $P(E) \geq p > 0$ implies $\text{KL}(P\|Q) = +\infty$, contradicting (55). Thus $Q(E) > 0$. By the data processing inequality,

$$\text{KL}(P\|Q) \geq \text{kl}(P(E), Q(E)).$$

For $a \in [0, 1]$ and $b \in (0, 1)$, with the usual endpoint conventions,

$$\text{kl}(a, b) = -h(a) - a \log b - (1 - a) \log(1 - b) \geq a \log \frac{1}{b} - \log 2,$$

where $h(a) \leq \log 2$ is the binary entropy. Therefore,

$$\text{KL}(P\|Q) > p \log \frac{1}{\delta} - \log 2.$$

The restriction $\delta \leq 2^{-2/p}$ implies $\log 2 \leq (p/2) \log(1/\delta)$, and hence

$$\text{KL}(P\|Q) > \frac{p}{2} \log \frac{1}{\delta},$$

contradicting (55). $\square$

Lemma 18. If $X \leq M$ almost surely and $\mathbb{E}X \geq \mu > 0$, then

$$\mathbb{P}\left(X \geq \frac{\mu}{2}\right) \geq \frac{\mu}{2M}.$$

If X is square integrable and $\mathbb{E}X > 0$, then

$$\mathbb{P}\left(X \geq \frac{1}{2}\mathbb{E}X\right) \geq \frac{(\mathbb{E}X)^2}{4\mathbb{E}[X^2]}.$$

Proof. For the first claim,

$$\mathbb{E}X \leq \frac{\mu}{2} + M\mathbb{P}(X \geq \mu/2).$$

For the second claim, putting $a := \mathbb{E}X/2$ and using Cauchy–Schwarz gives

$$\mathbb{E}X \leq a + \mathbb{E}[X\mathbf{1}\{X \geq a\}] \leq a + \sqrt{\mathbb{E}[X^2]\mathbb{P}(X \geq a)}.$$

Rearranging proves both statements. $\square$

Lemma 19. *Let $S_1, \dots, S_r$ be measurable functions with values in $[-M, M]$ on a common measurable space, and let $P, Q_1, \dots, Q_r$ be probability measures on that space. Put*

$$\bar{Q} := \frac{1}{r} \sum_{j=1}^r Q_j.$$

Then

$$\mathbb{E}_P \max_{j \in [r]} S_j \geq \frac{1}{r} \sum_{j=1}^r \mathbb{E}_{Q_j} S_j - 2M \cdot \text{TV}(\bar{Q}, P).$$

Proof. Since $\max_j S_j \geq S_i$ for every i ,

$$\mathbb{E}_{\bar{Q}} \max_j S_j = \frac{1}{r} \sum_{i=1}^r \mathbb{E}_{Q_i} \max_j S_j \geq \frac{1}{r} \sum_{i=1}^r \mathbb{E}_{Q_i} S_i.$$

The function $\max_j S_j$ takes values in $[-M, M]$, so its expectations under P and $\bar{Q}$ differ by at most $2M \cdot \text{TV}(\bar{Q}, P)$. $\square$

D.2 The action-count term when n/k is large

Proposition 2. *Let $1 \leq k \leq n/2$, let $r := \lfloor n/k \rfloor \geq r_\star = 2^{32}$, and suppose that*

$$T \geq 64nk \log \frac{en}{k}. \quad (56)$$

For every randomized policy on $\mathcal{X}_{n,k}$, there exists a deterministic adaptive non-anticipating loss process with every action loss in $[0, 1]$ such that

$$\mathbb{P} \left(R_T \geq 2^{-31} \sqrt{nT \log \binom{n}{k}} \right) \geq 2^{-10}. \quad (57)$$

Proof. Set

$$a := 2^{-6}, \quad \rho := 2^{-36}, \quad n' := kr, \quad B := \lfloor \log r \rfloor.$$

Since $r \geq 2^{32}$, we have $B \geq 1$. Define

$$x := \frac{T}{rB}, \quad \ell := \lfloor x \rfloor, \quad q := r\ell, \quad c_0 := 8\ell, \quad T_0 := qB.$$

For $b \in [B]$, let

$$I_b := \{(b-1)q + 1, \dots, bq\}.$$

We call the sets I_b the blocks, and for $t \in [T_0]$ let $b(t)$ be the unique index such that $t \in I_{b(t)}$. Because $n \geq kr$ and $B \leq \log r \leq \log(en/k)$,

$$x \geq 64k^2.$$

Thus $\ell \geq x/2$, and consequently

$$\frac{T}{2} \leq T_0 \leq T, \quad c_0 \geq 256k^2.$$

Set

$$\eta := k\sqrt{\frac{\rho}{c_0}}.$$

Then

$$\eta \leq \frac{\sqrt{\rho}}{16} < \frac{1}{16} = u_0.$$

We use the construction below during the first T_0 rounds and put the loss vector equal to zero afterward.

Independently draw

$$U_{b,i} \sim \text{Unif}[-1, 1] \quad (b \in [B], i \in [n']), \quad Z_t \sim \text{Unif}[-1, 1] \quad (t \in [T_0]).$$

We realize the learner's private randomization by an independent sequence $R = (R_t)_{t=1}^{T_0}$ of uniform seeds, one per round. Let P denote the joint law of (U, Z, R) ; in particular, the auxiliary table (U, Z) is independent of R . At round t , the learner's action is realized as a measurable function of $\mathcal{H}_{t-1}$ and R_t . Coordinates in $[n']$ are called signal coordinates; the remaining $n - n'$ coordinates are fillers. Within each block, for $i \in [n']$ define

$$C_{t,i} := \sum_{s < t: s \text{ in the same block}} \mathbf{1}\{i \in X_s\}, \quad A_{t,i} := \mathbf{1}\{C_{t,i} < c_0\}.$$

For fillers, set $A_{t,i}U_{b,i} := 0$. In block $b = b(t)$, let

$$\ell_{t,i} := \frac{1 + Z_t}{2k} - \frac{\eta}{k}g(Z_t)A_{t,i}U_{b,i}. \quad (58)$$

For any action $x \in \mathcal{X}_{n,k}$, put

$$h_t(x) := \sum_{i \in x} A_{t,i}U_{b(t),i}, \quad u_t(x) := -\frac{\eta}{k}h_t(x).$$

Then

$$\langle x, \ell_t \rangle = \frac{1}{2} + F_{u_t(x)}(Z_t).$$

Since $|u_t(x)| \leq \eta \leq u_0$, [Lemma 15](#) implies $\langle x, \ell_t \rangle \in [0, 1]$ for every action x . After the table (U, Z) is fixed, $A_{t,i}$ depends only on past actions, so the loss process is deterministic and non-anticipating. Along the realized action, abbreviate $u_t := u_t(X_t)$.

For every signal coordinate, define

$$S_i := \sum_{b=1}^B U_{b,i} \sum_{t \text{ in block } b} A_{t,i}g(Z_t),$$

and define the learner score

$$S_L := \sum_{t=1}^{T_0} g(Z_t) \sum_{i \in X_t} A_{t,i} U_{b(t),i}.$$

The common baseline in (58) cancels. Hence, using only comparators supported on the signal coordinates,

$$R_{T_0} \geq \frac{\eta}{k} \left(\max_{\substack{J \subseteq [n'] \\ |J|=k}} \sum_{i \in J} S_i - S_L \right). \quad (59)$$

The remainder of the proof establishes three estimates: a bound on the preference that the learner can create between the two signs of a hidden coordinate, a comparison between suitable mixtures of changed laws and the base law, and a positive average change in the comparator scores. They are stated in (60), (65), and (66), respectively, and are combined only at the end.

The learner's preference between the two signs. Fix a block-coordinate pair (b, i) . Condition on $|U_{b,i}| = u$, on all other U -coordinates, and on all auxiliary variables outside block b , but leave the Z -variables in block b and the learner's private seeds random. Write $\mathcal{G}_{b,i}$ for the sigma-field generated by the variables on which we condition. Compare the laws P_+ and P_- of the complete action-feedback history $\mathcal{H}_{T_0}$ corresponding to the signs $+u$ and $-u$. At a round at which i is not both selected and active, the feedback laws agree. At an active pull of i , their parameters differ by $2\eta u/k$. There are at most c_0 active pulls in the block. The sequential KL chain rule and (50) therefore give

$$\text{KL}(P_+ \| P_-) \leq 128c_0 \left(\frac{2\eta u}{k} \right)^2 = 512\rho u^2.$$

Pinsker's inequality yields

$$\text{TV}(P_+, P_-) \leq \sqrt{\frac{\text{KL}(P_+ \| P_-)}{2}} \leq 16u\sqrt{\rho}.$$

Consequently, for every $\sigma(\mathcal{H}_{T_0})$ -measurable random variable $V \in [0, M]$, for almost every realization of $\mathcal{G}_{b,i}$,

$$|\mathbb{E}[U_{b,i}V \mid \mathcal{G}_{b,i}]| = \frac{u}{2} |\mathbb{E}_+ V - \mathbb{E}_- V| \leq \frac{u}{2} M \text{TV}(P_+, P_-) \leq 8Mu^2\sqrt{\rho} \leq 8M\sqrt{\rho}.$$

Taking expectations gives

$$|\mathbb{E}[U_{b,i}V]| \leq 8M\sqrt{\rho}.$$

The variables

$$\sum_{t \text{ in block } b} A_{t,i} \quad \text{and} \quad \sum_{t \text{ in block } b} A_{t,i} \mathbf{1}\{i \in X_t\}$$

are $\sigma(\mathcal{H}_{T_0})$ -measurable and bounded by q and c_0 , respectively. Moreover, Z_t is independent of the past and of the current learner seed; hence it is conditionally independent of X_t given $\mathcal{H}_{t-1}$. Since $\mathbb{E}g(Z_t) = 2/3$,

$$Bq = T_0, \quad \frac{n'}{r} = k, \quad Bn'c_0 = 8kT_0.$$

Applying the preceding sign estimate with the two displayed random variables gives the first of the three promised estimates:

$$\frac{1}{r} \sum_{i=1}^{n'} \mathbb{E}S_i - \mathbb{E}S_L \geq -\frac{16}{3r} Bn'q\sqrt{\rho} - \frac{16}{3} Bn'c_0\sqrt{\rho} = -48kT_0\sqrt{\rho}. \quad (60)$$

Changes of variables that preserve the observed history. Define

$$T_a(u) := u + a(1 - u^2).$$

Since $a = 1/64$, the map T_a is an increasing diffeomorphism of $[-1, 1]$ onto itself and

$$T'_a(u) = 1 - 2au \geq \frac{31}{32}.$$

For $J \subseteq [n']$ with $1 \leq |J| \leq 2$, define a map Φ_J on the joint sample space of (U, Z, R) as follows. It leaves the learner seeds R unchanged and replaces $U_{b,j}$ by $T_a(U_{b,j})$ for every $j \in J$ and $b \in [B]$. The transformed feedback variables are constructed recursively. Suppose that the original and transformed histories agree through round $t - 1$. The common seed R_t then produces the same action X_t , and the activation variables at round t are also the same. Let u_t^J be the parameter computed from this action and the transformed hidden coordinates, and let Z_t^J be the unique solution of

$$F_{u_t^J}(Z_t^J) = F_{u_t}(Z_t).$$

This preserves the current feedback, so induction proves that Φ_J leaves the complete action-feedback history $\mathcal{H}_{T_0}$, and hence every activation variable, unchanged. Since T_a and every F_u involved here are increasing diffeomorphisms, the recursion can be run backwards: first apply T_a^{-1} to the changed hidden coordinates and then recover the original Z_t variables one round at a time. Thus Φ_J is a measurable bijection with a measurable inverse. Uniqueness of the recursion also gives

$$\Phi_{\{j,l\}} = \Phi_{\{j\}} \circ \Phi_{\{l\}} = \Phi_{\{l\}} \circ \Phi_{\{j\}} \quad (j \neq l).$$

Put

$$Q_J := P \circ \Phi_J^{-1}, \quad \Lambda_J := \frac{dQ_J}{dP},$$

For distinct j, l , abbreviate $\Phi_j := \Phi_{\{j\}}$, $\Phi_{jl} := \Phi_{\{j,l\}}$, $Q_j := Q_{\{j\}}$, $\Lambda_j := \Lambda_{\{j\}}$, $Q_{jl} := Q_{\{j,l\}}$, and $\Lambda_{jl} := \Lambda_{\{j,l\}}$. These derivatives exist: conditionally on the coordinates already generated, each changed coordinate is obtained through a one-dimensional diffeomorphism whose derivative is bounded away from zero, while the unchanged learner seeds cancel.

For later use, define

$$Y_t := F_{u_t}(Z_t), \quad m_{t,j} := A_{t,j} \mathbf{1}\{j \in X_t\}, \quad \delta_{t,j} := -\frac{a\eta}{k} m_{t,j} g(U_{b(t),j}).$$

Under Φ_j , the parameter at round t changes from u_t to $u_t + \delta_{t,j}$. Conditional on the past and the learner seed, the action is the same under every law being compared. Its probability therefore cancels from the likelihood ratio, leaving only the hidden-coordinate and feedback-density factors. Evaluated at the image of a base point,

$$\Lambda_j \circ \Phi_j = \prod_{b=1}^B \frac{1}{1 - 2aU_{b,j}} \prod_{t=1}^{T_0} \frac{p_{u_t}(Y_t)}{p_{u_t + \delta_{t,j}}(Y_t)}. \quad (61)$$

First,

$$\mathbb{E}_P[\Lambda_j^2] \leq \exp(10a^2 B). \quad (62)$$

Indeed,

$$\mathbb{E}_P[\Lambda_j^2] = \mathbb{E}_{Q_j} \Lambda_j = \mathbb{E}_P[\Lambda_j \circ \Phi_j].$$

Conditionally on the hidden U -table, the learner seeds, and the past, Y_t has density p_{u_t} . By (51), the conditional expectation of its factor in (61) is at most $e^{128\delta_{t,j}^2}$. In each block at most c_0 factors differ from one, and

$$\sum_{t \text{ in one block}} \delta_{t,j}^2 \leq \frac{a^2 \eta^2 c_0}{k^2} = a^2 \rho.$$

The hidden-coordinate factor satisfies

$$\mathbb{E} \frac{1}{1 - 2aU_{b,j}} = \frac{\operatorname{arctanh}(2a)}{2a} \leq \frac{1}{1 - 4a^2} \leq e^{8a^2}.$$

Integrating the temporal factors in chronological order, conditionally on U and R , bounds their product by $e^{128a^2\rho B}$. The B hidden-coordinate factors are independent and each has expectation at most e^{8a^2} . Therefore

$$\mathbb{E}_P[\Lambda_j^2] \leq e^{(8+128\rho)a^2 B} \leq e^{10a^2 B},$$

which proves (62).

For distinct signal coordinates j, l , define

$$N_{jl} := \frac{1}{k^2} \sum_{t=1}^{T_0} m_{t,j} m_{t,l}.$$

Take a base point and evaluate the likelihoods at its image under Φ_{jl} . At a round in which both coordinates are affected, the logarithm of the feedback-density contribution to $(\Lambda_j \Lambda_l / \Lambda_{jl}) \circ \Phi_{jl}$ is

$$H_{Y_t}(u_t + \delta_{t,j} + \delta_{t,l}) - H_{Y_t}(u_t + \delta_{t,l}) - H_{Y_t}(u_t + \delta_{t,j}) + H_{Y_t}(u_t).$$

All hidden-coordinate factors cancel. Consequently, (52) gives

$$\left| \log \left(\frac{\Lambda_j \Lambda_l}{\Lambda_{jl}} \circ \Phi_{jl} \right) \right| \leq 128a^2 \eta^2 N_{jl}.$$

Moreover,

$$N_{jl} \leq \frac{c_0 B}{k^2}, \quad \eta^2 N_{jl} \leq \rho B.$$

Since Φ_{jl} preserves the observed history and hence N_{jl} , change of variables and $e^x - 1 \leq xe^x$ give

$$\begin{aligned} \mathbb{E}_P[\Lambda_j \Lambda_l] - 1 &= \mathbb{E}_{Q_{jl}} \left[\frac{\Lambda_j \Lambda_l}{\Lambda_{jl}} \right] - 1 \\ &= \mathbb{E}_P \left[\frac{\Lambda_j \Lambda_l}{\Lambda_{jl}} \circ \Phi_{jl} \right] - 1 \\ &\leq 128a^2 \eta^2 e^{128a^2 \rho B} \mathbb{E}_P N_{jl}. \end{aligned} \tag{63}$$

A partition with few simultaneous selections. For every realization of $\mathcal{H}_{T_0}$,

$$\sum_{j \neq l} N_{jl} \leq T_0,$$

because at each round the number of ordered pairs of selected active signal coordinates is at most $k(k-1) < k^2$. Choose a uniformly random equipartition of $[n']$ into k groups of size r . A fixed ordered pair belongs to the same group with probability

$$\frac{r-1}{n'-1} \leq \frac{1}{k}.$$

Therefore, one deterministic partition $G_1, \dots, G_k$ satisfies

$$\sum_{g=1}^k \sum_{\substack{j,l \in G_g \\ j \neq l}} \mathbb{E}_P N_{jl} \leq \frac{T_0}{k}. \quad (64)$$

For this partition, define

$$Q_g := \frac{1}{r} \sum_{j \in G_g} Q_j, \quad \varepsilon_g := \text{TV}(Q_g, P).$$

For each group, direct expansion gives

$$\chi^2(Q_g \| P) = \frac{1}{r^2} \sum_{j,l \in G_g} (\mathbb{E}_P[\Lambda_j \Lambda_l] - 1).$$

Using (62), (63), and (64), the average off-diagonal contribution is at most

$$\frac{128a^2 \eta^2 e^{128a^2 \rho B} T_0}{kr^2} \frac{1}{k} = \frac{16a^2 \rho B}{r} e^{128a^2 \rho B}.$$

Thus

$$\frac{1}{k} \sum_{g=1}^k \chi^2(Q_g \| P) \leq \frac{e^{10a^2 B} - 1}{r} + \frac{16a^2 \rho B}{r} e^{128a^2 \rho B}.$$

Since $10a^2 < 1/32$ and $B \leq \log r$,

$$\frac{e^{10a^2 B} - 1}{r} \leq r^{-31/32} \leq 2^{-31},$$

while $16a^2 \rho = 2^{-44}$ and $128a^2 \rho = 2^{-41}$ give

$$\begin{aligned} \frac{16a^2 \rho B}{r} e^{128a^2 \rho B} &\leq 2^{-44} (\log r) r^{-1+2^{-41}} \\ &\leq 2^{-44} r^{-3/4+2^{-41}} \\ &\leq 2^{-67}. \end{aligned}$$

Here we used $\log r \leq r^{1/4}$ for $r \geq 2^{32}$. Thus the average chi-square divergence is at most 2^{-30} . The inequality $\text{TV}(Q, P) \leq \frac{1}{2} \sqrt{\chi^2(Q \| P)}$ and Jensen's inequality give

$$\frac{1}{k} \sum_{g=1}^k \varepsilon_g \leq \frac{1}{2} \sqrt{\frac{1}{k} \sum_{g=1}^k \chi^2(Q_g \| P)} \leq 2^{-16}. \quad (65)$$

The positive change in comparator scores. Fix a coordinate j and a block b . Put $\Delta U := ag(U_{b,j})$. At a round in which j is selected and active, the parameter changes by

$$s = -\frac{\eta}{k} \Delta U.$$

Since $|s| \leq a\eta/k \leq 1/1024$, the differential inequality (53) implies

$$|G_y(u + s) - G_y(u)| \leq 12|s|G_y(u).$$

Indeed, along the segment from u to $u + s$, $G_y(v) \leq e^{6|s|}G_y(u) \leq 2G_y(u)$. As $|T_a(U_{b,j})| \leq 1$, comparison of the old and changed score gives

$$\begin{aligned} S_j \circ \Phi_j - S_j &\geq a \sum_{b=1}^B g(U_{b,j}) \sum_{t \text{ in block } b} A_{t,j} g(Z_t) \\ &\quad - \frac{12a\eta}{k} \sum_{b=1}^B g(U_{b,j}) \sum_{t \text{ in block } b: j \in X_t} A_{t,j} g(Z_t). \end{aligned}$$

At within-block round τ , at most $k(\tau - 1)/c_0$ signal coordinates are inactive. Since $q = rc_0/8$,

$$\sum_{b=1}^B \sum_{t \text{ in block } b} \#\{i \in [n'] : A_{t,i} = 0\} \leq B \sum_{\tau=1}^q \frac{k(\tau - 1)}{c_0} \leq \frac{Bkq^2}{2c_0} = \frac{n'T_0}{16}.$$

Moreover, pathwise,

$$\sum_{i=1}^{n'} A_{t,i} g(U_{b,i}) \geq \sum_{i=1}^{n'} g(U_{b,i}) - \#\{i \in [n'] : A_{t,i} = 0\}.$$

Using $\mathbb{E}g(U) = \mathbb{E}g(Z) = 2/3$, summing the changed-score inequality over j , and using the preceding inactive-time bound, we obtain the third promised estimate:

$$\begin{aligned} \frac{1}{r} \sum_{j=1}^{n'} \mathbb{E}[S_j \circ \Phi_j - S_j] &\geq \frac{2a}{3r} \left(\frac{2}{3} n'T_0 - \frac{1}{16} n'T_0 \right) - \frac{12a\eta}{r} T_0 \\ &= \frac{29a}{72} kT_0 - \frac{12a\eta}{r} T_0 \\ &\geq \frac{a}{3} kT_0. \end{aligned} \tag{66}$$

For the last inequality, $29/72 \geq 3/8$, while $\eta \leq 1/16$, $k \geq 1$, and $r \geq 2^{32}$ imply $12\eta/r \leq k/24$.

Completion of the sparse action-count proof. For a group G_g , let

$$M_g := \max_{j \in G_g} S_j.$$

Define

$$\Gamma := \sum_{g=1}^k M_g - S_L.$$

Since $S_j \in [-T_0, T_0]$, [Lemma 19](#) gives

$$\mathbb{E}_P M_g \geq \frac{1}{r} \sum_{j \in G_g} \mathbb{E}_{Q_j} S_j - 2T_0 \varepsilon_g = \frac{1}{r} \sum_{j \in G_g} \mathbb{E}_P [S_j \circ \Phi_j] - 2T_0 \varepsilon_g.$$

Summing over the groups and using (60), (65), and (66), we find

$$\begin{aligned} \mathbb{E}_P \Gamma &\geq \left(\frac{a}{3} - 48\sqrt{\rho} - 2^{-15} \right) kT_0 \\ &\geq \frac{a}{4} kT_0. \end{aligned}$$

For the last inequality, note that $48\sqrt{\rho} + 2^{-15} = 7 \cdot 2^{-15} < a/12$. Also $\Gamma \leq 2kT_0$. Therefore, [Lemma 18](#) yields

$$\mathbb{P}\left(\Gamma \geq \frac{a}{8}kT_0\right) \geq \frac{a}{16} = 2^{-10}. \quad (67)$$

Since one may choose one maximizer from every group,

$$\max_{\substack{J \subseteq [n'] \\ |J|=k}} \sum_{i \in J} S_i \geq \sum_{g=1}^k M_g.$$

Thus, on the event in (67),

$$R_T = R_{T_0} \geq \frac{a\eta T_0}{8}. \quad (68)$$

The definitions give the exact identity

$$(\eta T_0)^2 = \frac{\rho}{8} n' k T_0 B.$$

Because $n' \geq n/2$, $T_0 \geq T/2$, and $B \geq (\log r)/2$,

$$\eta T_0 \geq \frac{\sqrt{\rho}}{8} \sqrt{nkT \log r}.$$

Furthermore, $n/k < r + 1$ and $r \geq 2^{32}$ imply

$$\log \binom{n}{k} \leq k \log \frac{en}{k} \leq 2k \log r.$$

Hence

$$\eta T_0 \geq \frac{\sqrt{\rho}}{16} \sqrt{nT \log \binom{n}{k}}.$$

Substituting $a = 2^{-6}$ and $\sqrt{\rho} = 2^{-18}$ into (68) gives the threshold $2^{-31} \sqrt{nT \log \binom{n}{k}}$. Finally, since (U, Z) is independent of the learner seeds R under the base law P , [Lemma 13](#) fixes only the table (U, Z) , without decreasing the probability in (67). $\square$

D.3 The action-count term when n/k is bounded

Proposition 3. *Let $1 \leq k \leq n/2$, assume $\lfloor n/k \rfloor < r_\star = 2^{32}$, and let*

$$T \geq n^2. \quad (69)$$

For every randomized policy on $\mathcal{X}_{n,k}$, there exists a deterministic loss sequence fixed in advance, with every action loss in $[0, 1]$, such that

$$\mathbb{P}\left(R_T \geq 2^{-38} \sqrt{nT \log \binom{n}{k}}\right) \geq 2^{-14}. \quad (70)$$

Proof. Construction and conditional gap. Put $D := 2k$ and use the first D coordinates, arranged into k pairs. For $\theta \in \{-1, +1\}^k$, set

$$\varepsilon := \frac{k}{8\sqrt{T}}. \quad (71)$$

The horizon condition and $k \leq n/2$ imply $\varepsilon \leq 1/16$. Define a probability distribution on $[D]$ by

$$p_\theta(2g-1) := \frac{1-\varepsilon\theta_g}{D}, \quad p_\theta(2g) := \frac{1+\varepsilon\theta_g}{D}.$$

At every round, independently draw $J_t \sim p_\theta$ and set

$$\ell_t := \frac{1}{3k}\mathbf{1} + \frac{2}{3}\left(e_{J_t} - \frac{1}{D}\mathbf{1}_{[D]}\right), \quad (72)$$

where $\mathbf{1}$ is the all-one vector in $\mathbb{R}^n$ and $\mathbf{1}_{[D]}$ is the indicator of the first D coordinates. For an exact k -set A , put

$$S := A \cap [D], \quad h := |S|,$$

and

$$z_g(S) := \mathbf{1}\{2g-1 \in S\} - \mathbf{1}\{2g \in S\}.$$

Then

$$\langle A, \ell_t \rangle = \frac{1}{3} + \frac{2}{3}\left(\mathbf{1}\{J_t \in S\} - \frac{h}{D}\right) \in [0, 1]. \quad (73)$$

Indeed, if $J_t \notin S$, the right-hand side is $1/3 - 2h/(3D) \geq 0$; if $J_t \in S$, it is $1 - 2h/(3D) \leq 1$. The other two bounds follow from $0 \leq h \leq D/2$.

For each θ , let $\mathbb{P}_\theta$ and $\mathbb{E}_\theta$ denote probability and expectation under the full joint construction. Realize all the learner's private randomization by one master seed $\mathbf{R}$, independent of $J_1, \dots, J_T$, and define

$$\mathcal{G}_0 := \sigma(\mathbf{R}), \quad \mathcal{G}_t := \sigma(\mathbf{R}, J_1, \dots, J_t), \quad t \in [T].$$

Then X_t and $S_t := X_t \cap [D]$ are $\mathcal{G}_{t-1}$ -measurable, while J_t is independent of $\mathcal{G}_{t-1}$. Let

$$P_\theta := \mathbb{P}_\theta^{\mathcal{H}^T}$$

be the induced law of the action-feedback history. Let x^θ select $2g-1$ from pair g when $\theta_g = +1$, and $2g$ when $\theta_g = -1$; this is the less likely item in the pair. Define

$$h_\theta(S) := k - \sum_{g=1}^k \theta_g z_g(S), \quad \Delta_t^\theta := \langle X_t, \ell_t \rangle - \langle x^\theta, \ell_t \rangle, \quad a_t^\theta := \mathbb{E}_\theta[\Delta_t^\theta \mid \mathcal{G}_{t-1}].$$

Since S_t is fixed conditionally on $\mathcal{G}_{t-1}$, a direct calculation gives the identity that will be used for both the mean and the second moment:

$$a_t^\theta = \frac{2\varepsilon}{3D} h_\theta(S_t) = \frac{\varepsilon}{3k} h_\theta(S_t) \in \left[0, \frac{2\varepsilon}{3}\right]. \quad (74)$$

Information contained in the feedback. Given S_t , the scalar feedback in (73) is an invertible affine function of the Bernoulli variable $\mathbf{1}\{J_t \in S_t\}$, whose parameter is

$$\pi_\theta(S) = \frac{h}{D} - \frac{\varepsilon}{D} \sum_{g=1}^k \theta_g z_g(S). \quad (75)$$

Let $\theta^{(g)}$ be obtained from θ by flipping coordinate g . If pair g is not split by S , the two Bernoulli parameters agree. If it is split, their difference has absolute value $2\varepsilon/D$. Let $s(S)$ be the number of split pairs. Since $h \leq D/2$, we have $s(S) \leq h$. When $h \geq 1$, both parameters satisfy

$$\pi_{\theta^{(g)}}(S) \geq \frac{(1-\varepsilon)h}{D} \geq \frac{3h}{4D}, \quad 1 - \pi_{\theta^{(g)}}(S) \geq \frac{3}{8}.$$

Using $\text{kl}(a, b) \leq (a - b)^2 / (b(1 - b))$, and treating $h = 0$ separately, we obtain the uniform one-step estimate

$$\sum_{g=1}^k \text{kl}(\pi_\theta(S), \pi_{\theta^{(g)}}(S)) \leq \frac{16\varepsilon^2}{D}. \quad (76)$$

Indeed, each split pair contributes at most $128\varepsilon^2 / (9hD)$, and there are at most h such pairs. Given the same past history, the learner uses the same conditional action law under θ and $\theta^{(g)}$, so the action term in the sequential KL chain rule is zero. Conditional on the action, the observation term is exactly the Bernoulli divergence in (76). The chain rule therefore gives

$$\sum_{g=1}^k \text{KL}(P_\theta \| P_{\theta^{(g)}}) \leq \frac{16\varepsilon^2 T}{D} = \frac{8\varepsilon^2 T}{k}. \quad (77)$$

A random round and the mean comparator gap. Choose a round τ uniformly from $[T]$ and independent fair Rademacher signs $\xi_1, \dots, \xi_k$, all independently of the action–feedback history. Let ν be the joint law of $(\tau, \xi_1, \dots, \xi_k)$ and put $\bar{P}_\theta := P_\theta \otimes \nu$. The common independent factor does not change the divergences:

$$\text{TV}(\bar{P}_\theta, \bar{P}_{\theta^{(g)}}) = \text{TV}(P_\theta, P_{\theta^{(g)}}), \quad \text{KL}(\bar{P}_\theta \| \bar{P}_{\theta^{(g)}}) = \text{KL}(P_\theta \| P_{\theta^{(g)}}).$$

For every pair g , define an estimator $\hat{\theta}_g$ as follows. If X_τ contains exactly one member of pair g , output $+1$ when that member is $2g - 1$ and output -1 when it is $2g$; if it contains both or neither, output ξ_g . Thus $\hat{\theta}$ is a measurable function of $(\mathcal{H}_T, \tau, \xi_1, \dots, \xi_k)$, defined by the same rule under every hypothesis. Conditional on S_τ and θ , the error probability in pair g is exactly one half of the contribution of pair g to $h_\theta(S_\tau)$; explicitly,

$$\bar{P}_\theta \left(\hat{\theta}_g \neq \theta_g \mid \tau, S_\tau \right) = \frac{1}{2} (1 - \theta_g z_g(S_\tau)).$$

Summing over g and averaging over τ therefore gives

$$\mathbb{E}_{\bar{P}_\theta} d_H(\hat{\theta}, \theta) = \frac{1}{2T} \mathbb{E}_\theta \sum_{t=1}^T h_\theta(S_t). \quad (78)$$

For every g , pair the hypotheses θ and $\theta^{(g)}$. If $B_{\theta, g} := \{\hat{\theta}_g = \theta_g\}$, then

$$\bar{P}_\theta(B_{\theta, g}^c) + \bar{P}_{\theta^{(g)}}(B_{\theta, g}) \geq 1 - \text{TV}(\bar{P}_\theta, \bar{P}_{\theta^{(g)}}) = 1 - \text{TV}(P_\theta, P_{\theta^{(g)}}).$$

Summing this inequality over all hypotheses θ and coordinates g gives

$$2^{-k} \sum_{\theta} \mathbb{E}_{\bar{P}_\theta} d_H(\hat{\theta}, \theta) \geq \frac{k}{2} - 2^{-(k+1)} \sum_{\theta} \sum_{g=1}^k \text{TV}(P_\theta, P_{\theta^{(g)}}).$$

For each fixed θ , Pinsker and Cauchy–Schwarz, followed by (77), give

$$\sum_{g=1}^k \text{TV}(P_\theta, P_{\theta^{(g)}}) \leq \sqrt{\frac{k}{2} \sum_{g=1}^k \text{KL}(P_\theta \| P_{\theta^{(g)}})} \leq 2\varepsilon \sqrt{T}.$$

Hence

$$2^{-k} \sum_{\theta} \mathbb{E}_{\bar{P}_{\theta}} d_H(\hat{\theta}, \theta) \geq \frac{k}{2} - \varepsilon \sqrt{T} = \frac{3k}{8}. \quad (79)$$

Combining (78) and (79), there exists a θ such that

$$\mathbb{E}_{\theta} \sum_{t=1}^T h_{\theta}(S_t) \geq \frac{3kT}{4}.$$

For this θ , define the fixed-comparator gap

$$G_T^{\theta} := \sum_{t=1}^T \Delta_t^{\theta}.$$

Taking expectations and summing (74) gives

$$\mathbb{E}_{\theta} G_T^{\theta} = \frac{\varepsilon}{3k} \mathbb{E}_{\theta} \sum_{t=1}^T h_{\theta}(S_t) \geq \frac{\varepsilon T}{4} = \frac{k\sqrt{T}}{32}. \quad (80)$$

From the mean to a positive-probability event. Consequently, with

$$A_T^{\theta} := \sum_{t=1}^T a_t^{\theta}, \quad M_t^{\theta} := \sum_{s=1}^t (\Delta_s^{\theta} - a_s^{\theta}),$$

we have

$$0 \leq A_T^{\theta} \leq \frac{2\varepsilon T}{3} = \frac{k\sqrt{T}}{12},$$

and $(M_t^{\theta})_{t=0}^T$ is a martingale with respect to $(\mathcal{G}_t)_{t=0}^T$. Since every Δ_t^{θ} belongs to $[-1, 1]$,

$$\mathbb{E}_{\theta}[(M_T^{\theta})^2] = \sum_{t=1}^T \mathbb{E}_{\theta} \left[\text{Var}_{\theta}(\Delta_t^{\theta} \mid \mathcal{G}_{t-1}) \right] \leq T.$$

Since $G_T^{\theta} = A_T^{\theta} + M_T^{\theta}$, it follows that

$$\mathbb{E}_{\theta}[(G_T^{\theta})^2] \leq 2 \frac{k^2 T}{144} + 2T \leq 3k^2 T. \quad (81)$$

Since $R_T \geq G_T^{\theta}$ pathwise, applying the second part of Lemma 18 to (80)–(81) gives

$$\mathbb{P}_{\theta} \left(R_T \geq \frac{k\sqrt{T}}{64} \right) \geq \frac{1}{12288} > 2^{-14}. \quad (82)$$

Since $\lfloor n/k \rfloor < 2^{32}$, we have $k > 2^{-32}n$. Also $\log \binom{n}{k} \leq n \log 2 < n$. Thus

$$\frac{k\sqrt{T}}{64} > 2^{-38} \sqrt{nT \log \binom{n}{k}}.$$

Finally, with θ already fixed, apply Lemma 13 to the table $\Xi = (J_1, \dots, J_T)$, which is independent of the learner's master seed. This produces a deterministic loss sequence fixed in advance and satisfying (70). $\square$

The preceding two action-count constructions can now be used through a single statement.

Corollary 2. *Let $1 \leq k \leq n/2$ and suppose that*

$$T \geq \max \left\{ n^2, 64nk \log \frac{en}{k} \right\}.$$

For every randomized policy on $\mathcal{X}_{n,k}$, there exists a deterministic adaptive non-anticipating loss process with every action loss in $[0, 1]$ such that

$$\mathbb{P} \left(R_T \geq 2^{-38} \sqrt{nT \log \binom{n}{k}} \right) \geq 2^{-14}.$$

Proof. If $\lfloor n/k \rfloor \geq r_*$, apply [Proposition 2](#); its threshold 2^{-31} and probability 2^{-10} are both stronger than the displayed ones. If $\lfloor n/k \rfloor < r_*$, apply [Proposition 3](#). $\square$

D.4 The confidence term

Proposition 4. *Let $1 \leq k \leq n/2$, let $0 < \delta \leq 2^{-36}$, and suppose that*

$$T \geq 4nk^2 \log \frac{1}{\delta}. \quad (83)$$

For every randomized policy on $\mathcal{X}_{n,k}$, there exists a deterministic adaptive non-anticipating loss process with every action loss in $[0, 1]$ such that

$$\mathbb{P} \left(R_T \geq 2^{-17} \sqrt{nT \log \frac{1}{\delta}} \right) \geq \delta. \quad (84)$$

Proof. Construction and the first case. Fix a core $C \subset [n]$ with $|C| = k - 1$, and put

$$V := [n] \setminus C, \quad N := |V| = n - k + 1 \geq \frac{n}{2}.$$

For an action X_t , write $h_t := |X_t \cap V|$. Since X_t has size k and C has size $k - 1$, we have $1 \leq h_t \leq k$. Define

$$p_t := \frac{h_t - 1}{8k}, \quad C_t := \sum_{s=1}^t p_s, \quad C_0 := 0.$$

Thus $0 \leq p_t < 1/8$ and $(C_t)_{t=0}^T$ is nondecreasing. Set

$$c_1 := 2^{-12}, \quad \Delta := c_1 \sqrt{\frac{NL_\delta}{T}}, \quad B := \frac{\Delta T}{16}, \quad A_t := \mathbf{1}\{C_{t-1} < B\}. \quad (85)$$

The horizon condition gives

$$k\Delta \leq \frac{c_1}{2}, \quad \Delta \leq 2^{-13}. \quad (86)$$

In particular,

$$4\Delta \leq 2^{-11} < u_0,$$

so the density estimates used below apply.

Under the reference construction P_0 , draw independent $Z_t \sim \text{Unif}[-1, 1]$ and define

$$\ell_{t,j}^{(0)} := \frac{1/4 + Z_t/8}{k} + \frac{1}{8k} \mathbf{1}\{j \in V\}.$$

For a candidate $i \in V$, define the environment P_i in which i is favored by

$$\ell_{t,j}^{(i)} := \ell_{t,j}^{(0)} - \Delta g(Z_t) A_t \mathbf{1}\{j = i\}. \quad (87)$$

For any exact k -set with h items in V , its action loss under P_0 equals

$$\frac{1}{4} + \frac{Z_t}{8} + \frac{h}{8k} \in \left[\frac{1}{8} + \frac{1}{8k}, \frac{1}{2} \right].$$

Together with (86), this shows that all action losses under P_0 and P_i belong to $[0, 1]$.

The action $C \cup \{i\}$ is a valid comparator, and direct subtraction gives

$$R_T \geq \mathcal{R}_i := C_T + \Delta \sum_{t=1}^T g(Z_t) A_t \mathbf{1}\{i \notin X_t\}. \quad (88)$$

Under P_0 , regret is at least C_T . If

$$P_0(C_T \geq B) \geq \delta,$$

then P_0 already proves the result because

$$B = \frac{c_1}{16} \sqrt{NTL_\delta} \geq 2^{-17} \sqrt{nTL_\delta}.$$

In this branch, Lemma 13 fixes the sequence (Z_t) and produces the required deterministic adversary. Hence assume from now on that

$$P_0(C_T \geq B) < \delta. \quad (89)$$

Many candidates under the reference law. Under P_0 , the variables $g(Z_t)$ are independent, belong to $[0, 1]$, and have mean $2/3$. Hoeffding's inequality gives

$$P_0 \left(\sum_{t=1}^T g(Z_t) < \frac{T}{2} \right) \leq e^{-T/18} \leq \frac{1}{32},$$

where the final inequality follows from (83) and $\delta \leq 2^{-36}$. Therefore, by (89), the event

$$\mathcal{G} := \left\{ C_T < B, \quad \sum_{t=1}^T g(Z_t) \geq \frac{T}{2} \right\} \quad (90)$$

has probability at least $15/16$, because

$$P_0(\mathcal{G}) \geq 1 - \delta - \frac{1}{32} \geq \frac{15}{16}.$$

On $\mathcal{G}$, we have $A_t = 1$ at every round. Averaging (88) over $i \in V$ gives

$$\begin{aligned} \frac{1}{N} \sum_{i \in V} \mathcal{R}_i &= C_T + \Delta \sum_{t=1}^T g(Z_t) \left(1 - \frac{h_t}{N} \right) \\ &= \Delta \sum_{t=1}^T g(Z_t) \left(1 - \frac{1}{N} \right) + \sum_{t=1}^T \left(p_t - \frac{\Delta g(Z_t)(h_t - 1)}{N} \right). \end{aligned}$$

By (86), $1/(8k) \geq \Delta/N$, and hence every summand in the second line is nonnegative. Since $N \geq 2$, on $\mathcal{G}$ we have

$$\frac{1}{N} \sum_{i \in V} \mathcal{R}_i \geq \frac{\Delta T}{4}. \quad (91)$$

Also, on $\mathcal{G}$,

$$0 \leq \mathcal{R}_i < B + \Delta T = \frac{17}{16} \Delta T.$$

For each outcome, define

$$\alpha := \frac{1}{N} \left| \left\{ i \in V : \mathcal{R}_i \geq \frac{\Delta T}{8} \right\} \right|,$$

and, for each $i \in V$, let

$$E_i := \left\{ \mathcal{R}_i \geq \frac{\Delta T}{8} \right\}.$$

On $\mathcal{G}$, (91) and the upper bound on $\mathcal{R}_i$ give

$$\frac{1}{4} \leq \frac{1}{N \Delta T} \sum_{i \in V} \mathcal{R}_i \leq \alpha \frac{17}{16} + (1 - \alpha) \frac{1}{8},$$

and therefore $\alpha \geq 2/15$. Consequently,

$$\frac{1}{N} \sum_{i \in V} P_0(E_i) = \mathbb{E}_0 \alpha \geq \frac{2}{15} P_0(\mathcal{G}) \geq \frac{2}{15} \frac{15}{16} = \frac{1}{8}. \quad (92)$$

Choosing one candidate. Let

$$W_i := \sum_{t=1}^T A_t \mathbf{1}\{i \in X_t\}.$$

While $A_t = 1$, the cumulative penalty can exceed B by at most one increment, and $p_t < 1/8$. Therefore, pathwise,

$$\begin{aligned} \sum_{i \in V} W_i &= \sum_{t=1}^T A_t h_t = \sum_{t=1}^T A_t + 8k \sum_{t=1}^T A_t p_t \\ &\leq T + 8kB + k. \end{aligned} \quad (93)$$

Apply Lemma 16 with $p = 1/8$. There exists a candidate $i \in V$ such that

$$P_0(E_i) \geq \frac{1}{16}, \quad \mathbb{E}_0 W_i \leq \frac{16}{N} (T + 8kB + k). \quad (94)$$

Comparing the two environments. The laws $P_0^{\mathcal{H}^T}$ and $P_i^{\mathcal{H}^T}$ of the action–feedback history under the two environments differ only at active rounds on which i is selected. At such a round, after subtracting $1/4 + h_t/(8k)$ and multiplying by four, the observation under P_0 is $F_0(Z_t)$ and the observation when i is favored is $F_{-4\Delta}(Z_t)$. Since $4\Delta < u_0$, the sequential KL chain rule and (50) apply. Also,

$$\frac{k\Delta}{2} \leq 2^{-14}, \quad \frac{k}{T} \leq \frac{1}{8}, \quad 1 + \frac{k\Delta}{2} + \frac{k}{T} < 2.$$

Here the second inequality follows from $T \geq 4nk^2L_\delta$, $n \geq 2k$, and $L_\delta > 1$. Therefore,

$$\begin{aligned}
\text{KL}(P_0^{\mathcal{H}_T} \| P_i^{\mathcal{H}_T}) &\leq 128(4\Delta)^2 \mathbb{E}_0 W_i \\
&\leq 2^{15} \frac{\Delta^2}{N} (T + 8kB + k) \\
&= 2^{15} \frac{\Delta^2 T}{N} \left(1 + \frac{k\Delta}{2} + \frac{k}{T}\right) \\
&\leq 2^{16} c_1^2 L_\delta = 2^{-8} L_\delta.
\end{aligned} \tag{95}$$

The event E_i is $\sigma(\mathcal{H}_T)$ -measurable under both laws. Indeed, on a round with $i \notin X_t$, under both P_0 and P_i ,

$$Z_t = 8 \left(Y_t - \frac{1}{4} - \frac{h_t}{8k} \right).$$

On a round with $i \in X_t$, the corresponding summand in $\mathcal{R}_i$ is zero. Hence $\mathcal{R}_i$, and therefore E_i , is the same measurable function of $\mathcal{H}_T$ under the two laws. Apply [Lemma 17](#) with $p = 1/16$. Since $2^{-8}L_\delta \leq (p/4)L_\delta$ and $\delta \leq 2^{-36} < 2^{-32} = 2^{-2/p}$, we obtain

$$P_i(E_i) \geq \delta.$$

On E_i ,

$$R_T \geq \frac{\Delta T}{8} = \frac{c_1}{8} \sqrt{NTL_\delta} \geq 2^{-16} \sqrt{nTL_\delta},$$

which is stronger than (84). Finally, [Lemma 13](#) fixes the sequence $(Z_t)_{t=1}^T$ and produces a deterministic adaptive non-anticipating adversary. $\square$

Proof of [Theorem 3](#). Put

$$s := \min\{m, d - m\}, \quad H := \log \binom{d}{s}, \quad L := \log \frac{1}{\delta}, \quad B := \max\{H, L\}.$$

By [Lemma 14](#), it is enough to prove the result for exact s -set policies. The threshold (47) implies the hypotheses of both [Corollary 2](#) and [Proposition 4](#), with $(n, k) = (d, s)$.

If $H \geq L$, [Corollary 2](#) and $\delta \leq 2^{-36} < 2^{-14}$ give

$$\mathbb{P} \left(R_T \geq 2^{-38} \sqrt{dT B} \right) \geq \delta.$$

If $L > H$, [Proposition 4](#) gives the same conclusion because

$$2^{-17} \sqrt{dT L} \geq 2^{-38} \sqrt{dT B}.$$

Thus the conclusion holds in all cases. Finally, $B \geq (H + L)/2$, so

$$2^{-38} \sqrt{dT B} \geq 2^{-39} \sqrt{dT (H + L)}.$$

Since $\binom{d}{s} = \binom{d}{m}$, this is exactly the claimed threshold, and [Lemma 14](#) gives the corresponding loss process when $m = d - s > s$. $\square$

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